

**Explainable and Responsible
Artificial Intelligence (XAI)
Experiments on Insurance
Data**

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Abstract

In the era of Artificial Intelligence (AI), the potential impact of new AI technologies for societies and various industries is extensive. Within the insurance sector, it is possible to observe various AI-based implementations from conversational AI to telematics motor insurance. Certainly, the majority of potential benefits of AI go well beyond only insurers and their customers. However, the use of AI raises some important questions. With the rise of AI, machine learning and other data-driven technologies, questions about how to ensure they are used ethically or responsibly in ways that enhance rather than diminish trust and customer confidence have arisen. Although there are new toolkits for the implementation of responsible and eXplainable AI (XAI), these ventures are still in their infancy. Various open source insurance datasets are available to investigate. It is possible to study the potential benefits and possible boundaries of toolkits like AI Fairness 360 or DALEX package. This project aims to explore the potential of available open source toolkits with an insurance dataset. The starting points for this project are the open source toolkit DALEX: Explainers for Complex Predictive Models in R, the insurance datasets within the R package `CASdatasets` and the website IBM Trustworthy AI Tools. The computational experiments conducted in this project might be useful for a giving further insight about how these methods can be useful in the insurance sector. In terms of selected machine learning algorithms, both the classical performance metrics and XAI-oriented tools are examined by using a motorcycle claim data set. The main findings of this study are discussed along with the pros and cons of the methods used to obtain them.

Keywords: explainable AI (XAI), Claim data, Non-life Insurance, Machine learning

Declaration

I declare that this thesis was composed by myself and that the work contained therein is my own, except where explicitly stated otherwise in the text.

(Richard Huang, Callum Purcell)

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Chapter 1

Introduction

“Interpretable Machine Learning refers to methods and models that make the behaviour and predictions of machine learning systems understandable to humans”

Christoph Molnar

The revenues generated by Artificial Intelligence (AI) in the insurance industry are projected to increase by 23% between 2019 and 2024 reaching \$3.4 billion. However, there are concerns about the appropriateness of using black-box AI models in insurance practices. Generally, eXplainable AI (XAI) involves enhancing current AI models by developing their transparency, interpretability and explicability, ultimately aiming to make AI models more understandable to humans [Adadi and Berrada, 2018], [Floridi et al., 2018]. We will use ‘interpretable’ or ‘understandable’ interchangeably. Transparency and explainability of AI models are core requirements to achieve impactful trustworthy AI in society [Maynard et al., 2022]. Trustworthiness is a fundamental principle in the insurance sector, and the level of trust among insurance companies and their stakeholders can be directly influenced by improved explanations provided by XAI. The end-insurance consumers who lack trust and comprehension of the AI processes involved will exhibit reduced confidence in the decisions made by AI-enhanced systems [Burrell, 2016].

The special thing about XAI is that it is explainable. This means that it can clearly show how it came to the conclusions it reached. Often with AI tools such as Neural Networks you feed in a lot of data through a black-box model and a result is spat out without any explanation of where this result came from. This can be confusing and result in some people having less trust in what the AI is saying. Explainable AI is helpful since the paths the XAI takes to the results are clear and easy to understand. For example, if you feed an Explainable AI a picture of a duck it can firstly tell you that it is a duck and the unique part is that it will point to the beak or webbed feet as the reason why the image is a duck. This is where it differentiates itself from many forms of AI. Within the insurance sector, the XAI can help to point to which feature within a customer profile leads to a higher insurance premium. This can be useful for two reasons. Firstly, the company can be more confident when offering a lower or higher insurance premium. Secondly, they can justify the rate to the customer if the customer challenges them. They can point to the specific part of the customer’s data that leads to the higher rate of insurance.

In this project, we intend to use XAI to help us investigate an insurance dataset. We will be using the XAI tool called DALEX to help us look into these datasets. By showing that XAI can be used in insurance to obtain important results we argue that XAI can be more widely used and trusted. Within this project we will first carry out a literature review of similar projects. We will then outline our methodology that we will use on our insurance datasets.

This includes both instance level and dataset level methods and techniques. Finally we will provide our results obtained from analysing our insurance datasets. We will also discuss these results. Specifically we will mainly be trying to figure out which input features contribute the most to the final result of the predicted insurance rate, analysing both categorical and numerical variables. We will be using four models to analyse our dataset: Logistic Regression, Support Vector Classifier, Random Forest Classifier and Gradient Boosting Classifier. We will then conclude by summarising our findings and showing that XAI is a useful and trustworthy tool.

DALEX: moDeL Agnostic Language for Exploration and eXplanation

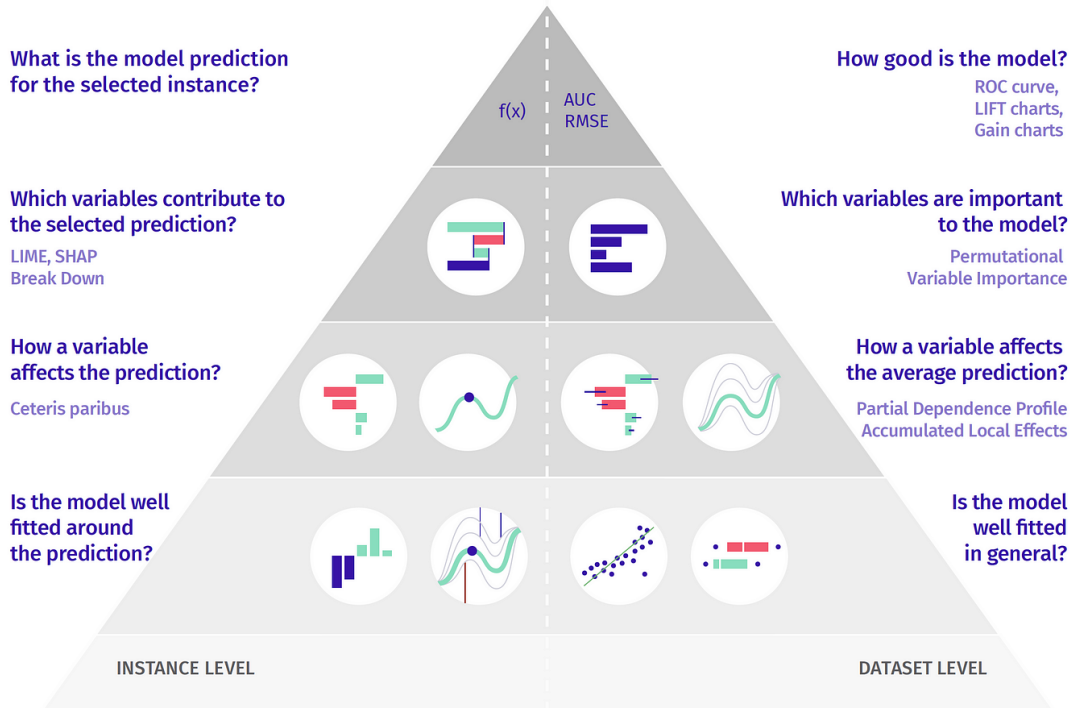


Figure 1.1: XAI explanation pyramid. Left side is about needs related to a single instance, right side to a model as a whole. Consecutive layers dig into more and more detailed questions about the model behaviour (local or global). Image Credit is [Biecek and Burzykowski, 2021]

In Figure 1.1, instance and dataset level tools are visualised together. For the Instance-level Exploration, this study mainly focuses on the Breakdown, Shapley values and Ceteris Paribus plot behaviours (left hand side), whereas for the Dataset-level Exploration, after looking at the overall model performance, this project relies on the use of variable importance, partial dependence profiles, local dependence plots and accumulated local effect plots. Additionally, not mentioned in 1.1, we opened a parenthesis on the fairness issue of the considered modelling tools.

From a practical standpoint, we primarily utilized the functionalities of the DALEX package for model exploration, alongside conventional model fitting procedures. One significant advantage of DALEX lies in its capability to establish a level of abstraction around a model, facilitating a more straightforward exploration and explanation of the model, as highlighted by [Biecek and Burzykowski, 2021]. Despite the existence of various other packages for the model building process, this particular package excels in seamlessly and consistently integrating model outputs into the DALEX ecosystem. This integration can be visually summarized in Figure 1.2.

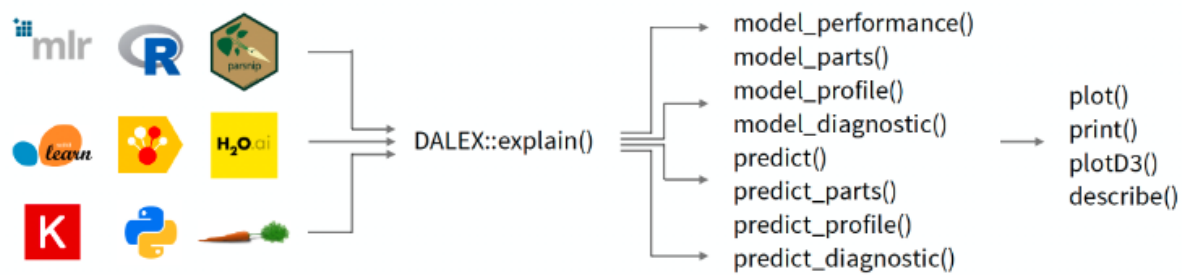


Figure 1.2: The key function is the `explain()` function, which wraps any model into a uniform interface. Then other functions from the DALEX package can be applied to the resulting object to explore the model. Image Credit is [Biecek and Burzykowski, 2021]

Chapter 2

Literature Review

“Today’s robots, like cleaning robots, robotic pets, or autonomous cars are far from being conscious enough to fall under Asimov’s ethics. However, we are more and more surrounded by complex predictive models and algorithms used for decision-making.”

Przemysław Biecek and Tomasz Burzykowski

AI and XAI

AI is becoming increasingly prevalent in our society and its implications for our future cannot be overstated. The reach of AI will be felt in almost all parts of our lives, from healthcare to motor insurance. AI and Machine Learning (ML) systems are becoming extremely accurate, and hence useful, in a number of different sectors. However, many of these methods are complicated and hard to understand. They are also what is described as “black-box”, meaning their inner workings and the methodology are not fully clear to an outside observer. This creates two problems. Firstly, if the ML model makes an error, it can be hard to discern where the error arose, due to the complicated nature of the model and the lack of transparency. Secondly, it can be hard to trust the model for this exact reason; it can make errors which may be obvious when a human looks at the conclusion. Since the model lacks transparency, it is difficult to amend this mistake. There is a solution; Explainable Artificial Intelligence (XAI). This is a special range of techniques which aim to aid data analysts among others to firstly increase their understanding about whatever dataset their modelling is based on and secondly to provide trustworthy conclusions which are transparent and interpretable.

Techniques of XAI

The main techniques used when talking about XAI are: feature interaction and importance, attention mechanism, dimensionality reduction, knowledge distillation and rule extraction and intrinsically interpretable models [Owens et al., 2022]. This paper will focus on feature importance, knowledge distillation and rule extraction. These techniques can be applied in a variety of circumstances. Many of the techniques are also model-agnostic, so can be applied to any model. We will use instance level techniques, such as breakdown plots, shapley values and ceteris paribus plots to firstly see how different variables affect the predictions our models make and which variables contribute the most to our selected prediction. We will then extend

this to the global dataset level and try to answer questions like which are the most important variables and how each variable affects the average prediction of the model.

XAI in healthcare

XAI is especially useful in healthcare in a number of scenarios; specifically medical insurance, prediction of illnesses and even of mortality. It has been proven that XAI can be useful in predicting all-cause mortality [Qiu et al., 2022]. Here, previously overlooked factors and interactions between features were identified and explained by XAI, allowing medical professionals to understand the data more easily. The XAI even allowed the conclusions to be easily understood by those without medical expertise or an understanding of the underlying ML models. The paper concluded that the "mortality risk scores" which the XAI helped to decide could help individuals improve self-awareness of their health status and help medical professionals identify patients with high risk. [Orji and Ukwandu, 2024] also outlined the importance of having an effective and transparent medical system. This is in part due to fairness reasons - a black box ML model could have unseen biases. These can be mitigated and hopefully fully removed with the use of XAI. XAI should make the job of humans easy, by showing exactly how it came to the conclusion it did, allowing for humans to easily block obviously incorrect decisions.

XAI in motor insurance

Finally we turn to XAI in motor insurance; as the focused application of this project. XAI is very effective in this sector because there are a number of variables all with varying effects on the insurance premium but which are not obvious to the human eye. [Lorentzen and Mayer, 2020] summarises tools for explaining black box ML models using real insurance datasets. This paper carries out many of the same techniques that we will be applying and concludes that XAI is becoming increasingly applicable when compared to classic statistical modelling techniques. [Mayer et al., 2023] dives deeper into one of the techniques used in feature importance - Shapley Additive Explanation (SHAP). This is a very important technique for our project and many XAI tools. SHAP analyses a model by trying to ascertain how much each of the features individually contribute to the overall output of the model. It does this by summing the impact of each feature. The project concludes that SHAP is an extremely effective and easy to implement technique to add to our explainability arsenal. This is because the decompositions made by SHAP can be analysed by simple graphical techniques, improving the interpretability. We can see that research into XAI is extensive and ongoing.

In conclusion, there has been a lot of research into XAI, nicely summarised in [Owens et al., 2022], where the authors analysed and classified 103 articles (between 2000-2021) to highlight the current state-of-the-art XAI in insurance. We can see that XAI is a rapidly growing idea and its implementation into a variety of industries like healthcare and insurance is becoming more and more common. The importance of XAI in the future cannot be understated and it will shape our lives in many ways in the years to come.

Chapter 3

Methodology

“... all models are approximations. Essentially, all models are wrong, but some are useful. ”

George Box

3.1 Modelling Tools

There are two main methods that we follow; explanatory analysis and predictive modelling. Explanatory analysis is used to analyse a dataset and explain the trends and observations that one can obtain from the data. However, predictive modelling is used to predict the future outcomes given the data. In this project, we will be using both approaches to digest characteristics of our dataset in depth. In terms of modelling part, firstly, we explain what the most important variables and features are and then to predict future outcomes based on the data we have. In terms of the problem, the main goal is to classify claim vs non-claim cases based on the selected data set, using the XAI tools from the DALEX ecosystem.

Logistic Regression

Our dependent variable is whether or not an insurance claim has occurred, which is binary. Hence our first model will be Logistic Regression as this is a simple way to model binary output data and we can use it as a benchmark to compare our other models against. Mathematically, it is an extension of the linear regression model for classification problems, by changing the underpinning distributional assumption from Normal to Bernoulli. The structural form of the logistic regression model over one single predictor is;

$$\text{logit}(E[Y_i|X_i]) = \text{logit}(p_i) = \ln\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_{1i} + \epsilon \quad (3.1)$$

where Y_i denotes the outcome (or dependent) variable, $X_{1,i}$ denotes the predictor of interest or the independent variable for subject i , β_0 is the intercept term, β_1 denotes the slope parameter and ϵ denotes the error terms. Specifically, this method relies on the logit link function that models the mean of the response as a linear combination of the given set of predictors. Link functions in Generalized Linear Models (GLMs) such as Logistic Regression connect the linear predictor of the model to the expected value of the response variable. Even if the above setting is given for single predictor case, the extension to the p -dimensional scenario looks similar.

The logistic regression model has several assumptions; not necessarily following the same assumptions as linear regression models. The logistic regression model employs maximum likelihood estimation (MLE), a method based on conditional probability that categorises the

outcome when a specific threshold is reached (e.g., > 0.50 , which is called the Bayes classifier boundary). Generally, the probability range of a logistic regression model is from 0 to 1, as a result of the logit link function. As mentioned in the main function above, the logit function is the natural log of the odds ($\ln(\frac{p_i}{1-p_i})$) that is modelled as a linear function of the given predictor.

These two models are not the only options when it comes to GLM models; there are a wide variety of link functions that make up GLMs. Considering the regularized version of logistic regression is useful when the feature space is higher dimensional and incorporating nonlinear basis extensions of the predictors might lead to increased accuracy, but both of these techniques would reduce interpretability. For the sake of simplicity, this project relies on the default logistic regression setting only. This will be used as a benchmark for comparisons of more complex models.

Support Vector Machines (SVMs)

The second model we will be using to investigate our dataset is Support Vector Machine classifier. This is a supervised learning method that aims to find a hyperplane that classifies the points. The hyperplane resides in the dimension space of the number of features. The main aim of SVMs is to find the hyperplane that creates the biggest separation between the datapoints.

Given any predictor matrix ($X_{n \times p}$) with a binary response variable $y \in \{-1, 1\}$, it might be possible to define a p -dimensional hyperplane geometrically,

$$h(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \cdots + \beta_p x_p = X'_i \beta + \beta_0 = 0 \quad (3.2)$$

so that all of the $y_i = -1$ observations fall on the negative side of the hyperplane and the $y_i = 1$ observations fall on the positive side, following the given inequality below;

$$y_i (x'_i \beta + \beta_0) > 0 \quad (3.3)$$

In this setting, this separating hyperplane is a simple classifier and the magnitude of $f(x_i) = (x'_i \beta + \beta_0)$ is an indicator of confidence in the predicted classification. A hyperplane's margin, $M = \min \{y_i(x'_i \beta + \beta_0)\}$, is the perpendicular distance to the closest observation. As a key concept of the SVM classifier, the maximal margin classifier is the hyperplane that maximizes the margin. The larger the total separation between the points, the more confident we can be that the SVM hyperplane is a good classifier. Behind the scenes, Support Vector Machines utilize the hinge loss function, which different from classical linear models.

We can also add a regularization parameter to the cost function to balance the loss function and maximising the distance between the hyperplane and the data points. SVMs are good with high dimensional data and when there is not a lot of data, however, they struggle with large datasets or when there is lot of noise in the data.

Decision Trees

We will describe the process of Decision tree classifiers with the aim of explaining how Random Forest algorithms work. Decision Trees (DTs) are a non-parametric supervised learning technique utilised for both classification and regression tasks. The main objective is to develop a model that can forecast the value of a specific variable by acquiring basic decision rules from the data's characteristics. In general, decision trees can be treated as a series of if-then-else choices to classify data into various subgroups.

This method segments the predictor space into K non-overlapping terminal nodes (leaves). Each node is described by a set of rules which can be used to predict new responses. It defines the nodes through a top-down greedy process called recursive binary splitting. The process is top-down as it initiates at the highest level of the hierarchy with all observations in one region and then progressively divides the predictor space. It is greedy because it selects the best split at each step without taking into account future splits. During that process, the best split is the

predictor variable and cutpoint that minimises a specified cost function. The most common cost function for the classification problem is the Gini index;

$$G = \sum_{c=1}^C \hat{p}_{kc}(1 - \hat{p}_{kc}), \quad (3.4)$$

where $\hat{p}_{kc}$ is the proportion of training observations in node k that are class c .

DTs can get to be very large and cumbersome if the maximum depth is not kept in check, however, decision tree classification is often a simple modelling technique that is easy to interpret and visualise. Decision tree models are also white box - we can easily see the inner workings of the model and make adjustments accordingly. This helps with error detection and improves the explainability of the model. Another benefit of using decision tree models is that no preprocessing is required and that they are also robust; they are good at handling different types of data. However, one drawback is that decision trees can be prone to overfitting if the hyperparameters like the maximum depth are not monitored carefully. It is possible to overcome such issues by increasing the complexity through models such as random forest modelling.

Random Forests (RFs)

One main drawback of decision trees is that they are high-variance estimators. A small number of additional training observations can dramatically change the prediction performance of a created decision tree. In order to eliminate this issue, one of the proposed methods is called bagging (bootstrap aggregation). This bagging algorithm constructs B regression trees using B bootstrapped training sets and averages the resulting predictions. However, these bootstrapped training observations can create another type of problem during the calculation, called correlated trees. Even if the aim of bagging is to reduce the variance, the correlated trees that rely on the bootstrapped samples can impact the model performance.

To combat this, Random Forests improve bagged trees by way of a small tweak that de-correlates the trees. While creating the trees, a random sample of available predictors is chosen as split candidates from the full set of, say, p predictors. This small tweak allows a fresh sample of predictors to be taken at each split, helping to de-correlate the trees. The random forest algorithm considers multiple decision trees that are all trained slightly differently (because of different subset of predictors). However, all of them are taken into consideration for the final classification. The main difference can be visualised in Figure 3.1.

Gradient Boosting Method (GB)

Various recently published studies propose improvements to the tree based models described above. Among these, one of the more commonly known algorithms is called the Gradient Boosting (GB) method, based on the boosting strategy and multiple tree based models. GB uses decision trees as base (weak) learners; combining many weak learners to form a strong learner. For this reason, it is called as an ensemble learning method; using the output of many models (i.e. trees) in the final prediction. GB models will sequentially create a series of weak models on different parts of the data, assigning a weight to each weak prediction based on the performance of the weak learner. At the end, the final prediction is calculated by summing the weighted values of all base learners. Tree-based boosting models are good at handling high-order interactions but can be prone to overfitting.

In conclusion, we will be using these four models - Logistic Regression, Support Vector Machine, Random Forests, and Gradient Boosting classifiers. This allows us to cast a wide net and cover our bases when investigating this dataset in case one of our models is not entirely accurate.

Additionally, throughout most case studies in this area, a common approach is to consider the logistic regression model as the benchmark and create more complex alternatives to im-

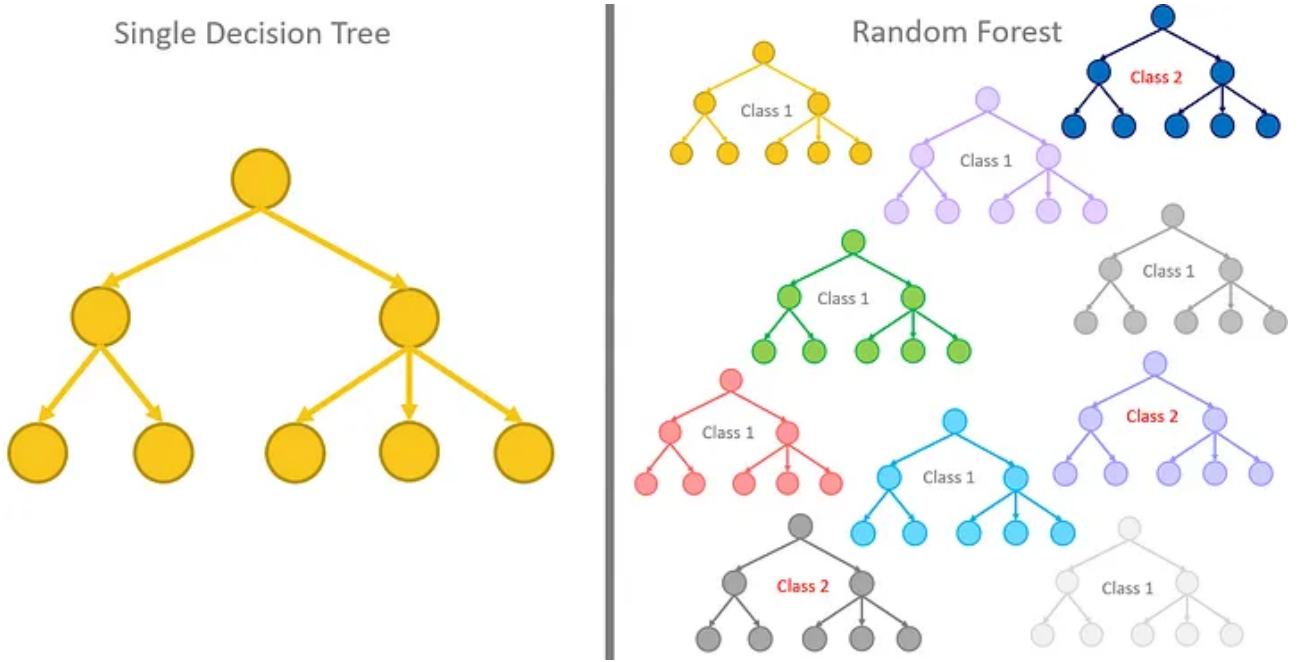


Figure 3.1: The difference between the DT and RF methods visually (Image Credit: From a Single Decision Tree to a Random Forest (Rosaria Silipo))

plement a model performance in order to compare to the baseline of logistic regression. We will consider various classification models and discuss not only the overall performance but also instance or dataset level differences between the models. This can bring further insights regarding the explainability with regards to the insurance sector.

3.2 Over Sampling Method

In various real life classification problems, a critical issue that one can encounter is the class proportionality. This phenomenon is described as the imbalanced data problem. In a simpler binary classification setting (i.e. in our case, claim or no claim), this problem can result in misleading model performance results which can eventually impact the explainable part of the project. Our dataset unfortunately has a proportionally small number of claims made when compared to instances without a claim made. Due to this lack of data, we have implemented a method to mitigate the effects of a small number of claims. The main focus of the method employed is to resample the data to adjust the distribution of the classes to be more equal so that the dataset is easier to analyse and make conclusions about.

Various methods have been proposed to resolve data imbalance issues before fitting classification models. Two widely known techniques are called Over-sampling and Under-sampling. The simplest implementation of over-sampling is to duplicate random records from the minority class relying on the resampling observations with replacement. On the other side, under-sampling involves randomly removing observations from the majority class, by the resampling without replacement. Beyond these two main approaches, further strategies include SMOTE (Synthetic Minority Over-sampling Technique) and NearMiss.

In this project, the random over-sampling approach is followed to increase the observations belonging to the claim cases. This increases the proportionality of claim numbers against non-claim numbers when compared to the original dataset. We will apply this method by using an R package named ROSE [Lunardon and Torelli, 2014]. This ROSE package offers a comprehensive solution to the challenges posed by imbalanced classification problems. By generating artificial balanced samples using a smoothed bootstrap approach, ROSE aids in both

model estimation and accuracy evaluation for binary classifiers operating in the presence of rare classes. This package provides us with versatile evaluation metrics tailored for imbalanced datasets, which allows for a more nuanced assessment of model performance. Additionally, we should be cautious of potential biases or overfitting issues introduced by the generation of artificial samples and we should thoroughly validate model performance.

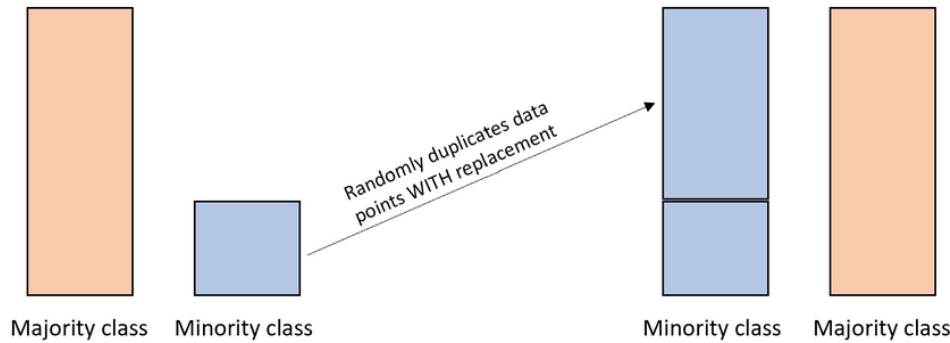


Figure 3.2: Random oversampling [Wongvorachan et al., 2023]

3.3 Instance Level

Firstly, we will be investigating our dataset by looking at specific instances within the dataset. We can analyse these instances using breakdown plots, shapley values and ceteris paribus profiles. Instance level refers to a single observation that our model has predicted. We can analyse specifically how much each feature contributed to this prediction as well as other interesting properties.

We can use breakdown plots to visualise the exact effect of each feature on the prediction of our particular model. Breakdown plots show how each feature positively or negatively influences the final prediction that the model makes for a particular instance. This can be a quick way of seeing which features are the most important and which features have a positive or negative effect on the prediction [Biecek and Burzykowski, 2021]. However, the size of the influence of each feature is also dependent on which order you consider the variables. Different orderings of explanatory variables produce different breakdown plots. This is why we need Shapley values.

Shapley values are a common way of measuring feature importance for a particular instance. This is due to the relative ease with which they are calculated and also because they are easy to visualise in a graph. They are calculated by averaging the effects of each explanatory variable over all of the possible orderings of the variables. This removes any variability due to the orderings that we would see in breakdown plots [Biecek and Burzykowski, 2021]. Hence we are left with final positive or negative overall impacts of each feature for one particular instance.

Another way to investigate instances is via ceteris paribus profiles. For this analysis technique, we choose an explanatory variable (it must be numeric, not categorical), then we keep all other variables in the instance the same and alter the explanatory variable over its domain and see how the model prediction changes. This helps us see the impact of the explanatory variable over its domain for that instance [Biecek and Burzykowski, 2021]. We can plot this on a graph and get a nice visualisation of the explanatory variable's influence. However, this only works for numeric data because we assume no relation between the categories in categorical data so we cannot have a continuous graph. Another drawback to CP profiles is that they do not take into account correlation or interactions between variables. This could lead to misleading results and we will discuss the impact of correlation between our variables later on in the project.

These three instance level techniques allow us to analyse a wide range of properties for each instance while also remaining easy to interpret and explain.

3.4 Dataset Level

We will be investigating our dataset secondly at the global level; reaching conclusions and seeing general tendencies within the data, building on our analysis from the instance level section. We will be using three main dataset level techniques: variable importance measures, partial dependence profiles and local dependence/accumulated-local effects plots. These are three common methods that allow us to see the influences of the features in various ways.

We will first be investigating variable importance via the dropout loss function. This involves removing one variable and seeing how much of an effect this has on the model performance. Naturally, the most important variables will have the biggest negative effect on the model performance when dropped, so we will decide which variables are the most important based on which variables have the highest dropout loss. This allows us to identify the most important variables at a dataset level, which can lead to further investigations into that specific feature. It also allows us to identify the less useful explanatory variables, which we could possibly remove to simplify the model hence reducing model complexity [Biecek and Burzykowski, 2021]. This technique is susceptible to changes in results when different orderings of variables are used, so we will mitigate this by averaging the effect of each variable over 10 random orderings. This will make the results more consistent and accurate.

Secondly, we will be looking at partial dependence (PD) profiles. We will be employing two types of partial dependence profiles; namely for categorical and numerical data. For categorical data, PD profiles average the effect of each category in the chosen variable while keeping all other variables in the model constant. This allows us to see the impact of the each category within the variable chosen. Averaging the effect over multiple instances removes the variability that comes with some of the instance level analysis and allows for a more general conclusion about the dataset. We will also be using PD profiles for numerical data. When PD profiles analyse numerical data, they use the same idea as CP profiles in that they examine the effect of the explanatory variable over its domain. However, as with the categorical data, PD profiles average the effect over multiple instances or even the whole dataset [Biecek and Burzykowski, 2021]. In our case, we will be averaging over 300 instances. To further dive deeper into the effect of each explanatory variable, we will also be providing PD profiles for some of our categorical variables, but split into each category within the variable. We will then plot all of the PD profiles for each of the categories on one graph and then we can see how the numerical data varies over its domain for each of the categories within one categorical variable. This could lead to some interesting conclusions or insights about the interaction between the variables. Speaking of interactions, PD profiles are often subject to influence by interactions or correlations between variables. This could make our results slightly misleading and is why we need to consider local dependence and accumulated-local effects plots as well.

Local dependence (LD) and accumulated-local effects (ALE) plots can be seen as an extension of PD profiles. PD profiles are calculated using the marginal distribution of the explanatory variable in question. LD plots instead use the conditional distribution of the explanatory variable, conditional on the other variables, hence taking into account their influence on the model's prediction. However, as argued by [Apley and Zhu, 2020], correlation between the variables will still impact the LD plots despite using the conditional distribution. This is why we must introduce ALE plots. ALE plots examine the difference in prediction of the model in a small range around each value. They use this difference to produce an estimate of the model's prediction at that point which does not take into account interactions between variables [Biecek and Burzykowski, 2021]. This is because it looks at the difference in prediction, not the value of the prediction. This removes the effect of interactions and hence provides us with a very useful way of graphing our model's predictions.

In conclusion, we can investigate our model at the dataset level in various ways, using feature importance measures as well as examining the impact of specific explanatory variables.

3.5 Fairness models

We will also be investigating the fairness issue of the implemented data analysis. Artificial Intelligence is known to sometimes have biases and we want to identify any biases that our analysis may have. This is essential for fairness and making sure some groups in society are not unfairly treated through external factors. Data analysis such as XAI and Machine Learning promises to give accurate and unbiased predictions based purely on data and removing unfair biases that exist in the real world.

Specifically, the focus is to evaluate the fairness of our models when classifying men against classifying women (i.e. potential gender bias). We want to see if our model is biased towards either one of these groups. If it is, we can possibly alter the model to be more fair and appropriate. We will choose a privileged group (male) and compare various fairness measures against women. We will consider 5 fairness metrics. They are calculated using the true positive (TP), false positive (FP), true negative (TN) and false negative (FN) predictions that our model made. The 5 metrics are:

- Equal opportunity ratio: $(TP/(TP + FN))$ with abbreviation TPR
- Predictive parity ratio: $(TP/(TP + FP))$ with abbreviation ACC
- Predictive equality ratio: $(FP/(FP + TN))$ with abbreviation PPV
- Accuracy equality ratio: $((TP + TN)/(TP + FP + TN + FN))$ with abbreviation FPR
- Statistical parity ratio: $((TP + TP)/(TP + FP + TN + FN))$ with abbreviation STP

In general, the ratios between scores of privileged and unprivileged metrics should be close to 1. The closer to 1 the ratio is, the fairer the model. We can add a range for the given quantities to allow for some variability, represented via $(\epsilon, 1/\epsilon)$ where by default $\epsilon = 0.8$. From the practical point of view, the given metrics are calculated for each subgroup, and then their scores are divided by the score of the privileged subgroup.

If two or more of these metrics are not in a specified range that, we will determine that the model is biased and further action can be taken. We can also use parity loss plots to visualise the bias metrics. Parity loss represents the difference in the fairness metrics; a higher value represents a more extreme suggestion of bias in the model. We can visualise these plots using stacked bar charts. We can also use ceteris paribus cutoff plots which show the effect of changing the cutoff for a specified subgroup, in our case for women.

Chapter 4

Case Study

“The problem is that a single metric, such as classification accuracy, is an incomplete description of most real-world tasks”

Finale Doshi-Velez and Been Kim

In this project, one insurance dataset is considered for further exploration. This case study is conducted using the Swedish Motorcycle claim data set (`swmotorcycle`), available online through the R package called `CASdatasets`.

4.1 Data Description

We will be considering a dataset of Swedish Motorcycle Insurance information. We have followed the same cleaning technique employed in [Wüthrich and Merz, 2023], which includes changing the `Gender`, `Area` and other features to factors, to allow for easier analysis. We have also renamed some of the categories to allow for easier comparison and less cluttering on our graphs. Finally, we removed all the data of people below 18 since they cannot use a motorcycle and above 70 due to lack of sufficient data. We have also removed data from vehicles over the age of 30 for the same reason. The last bit of cleaning we did was to ensure that the motorcycle was actually used. Finally, we dropped the `ClaimAmount` variable because we are investigating whether or not a claim happened, not the amount of the claim, and introduced a new variable called `ClaimMade`, which takes the value 1 if a claim was made and 0 if there was no claim. This allowed us to perform our classification techniques described in Section 3. Here are the variables that we will be investigating, as set out in [Wüthrich and Merz, 2023];

1. `OwnerAge` - the age of the owner of the motorcycle in $\{18, \dots, 70\}$.
2. `VehAge` - the age of the motorcycle in $\{0, \dots, 30\}$.
3. `Gender` - the gender of the owner of the motorcycle.
4. `Area` - 7 geographical Swedish zones being (1) central parts of Sweden’s three largest cities, (2) suburbs and middle-sized towns, (3) lesser towns except those in zones (5)-(7), (4) small towns and countryside except those in zones (5)-(7), (5) Northern towns, (6) Northern countryside, and (7) Gotland (Sweden’s largest island).
5. `RiskClass` - the “risk” of the motorcycle classes as determined through a calculation involving the engine power and the vehicle weight. $(\text{Engine power in kW} \times 100) / (\text{Vehicle weight in kg} + 75)$.

6. **BonusClass** - bonus-malus class of the motorcycle, entry level is BM1.
7. **ClaimMade** - binary output of whether or not an insurance claim was made (new variable we created).

Figure 4.1 shows the R code for the data cleaning process. Figure 4.2 shows the first 20 rows for the dataset.

```

1 library(CASdatasets)
2 data(swmotorcycle)
3 mcdata <- swmotorcycle
4 mcdata$Gender <- as.factor(mcdata$Gender)
5 mcdata$Area <- as.factor(mcdata$Area)
6 mcdata$Area <- factor(mcdata$Area, levels(mcdata$Area) [c(1,7,3,6,5,4,2)])
7 mcdata$Area <- c("Zone 1", "Zone 2", "Zone 3", "Zone 4", "Zone 5",
8                  "Zone 6", "Zone 7") [as.integer(mcdata$Area)]
9 mcdata$Area <- as.factor(mcdata$Area)
10 mcdata$RiskClass <- as.factor(mcdata$RiskClass)
11 mcdata$RiskClass <- factor(mcdata$RiskClass,
12                            levels(mcdata$RiskClass) [c(1,6,7,3,4,5,2)])
13 mcdata$RiskClass <- as.integer(mcdata$RiskClass)
14 mcdata$BonusClass <- as.integer(as.factor(mcdata$BonusClass))
15 #
16 mcdata <- mcdata[which(mcdata$OwnerAge >= 18),] # only minimal age 18
17 mcdata$OwnerAge <- pmin(70, mcdata$OwnerAge) # set maximal age 70
18 mcdata$VehAge <- pmin(30, mcdata$VehAge) # set maximal motorcycle age 30
19 mcdata <- mcdata[which(mcdata$Exposure > 0),] # only positive exposures

```

Figure 4.1: Code of Data Cleaning [Wüthrich and Merz, 2023]

OwnerAge	Gender	Area	RiskClass	VehAge	BonusClass	ClaimMade
18	Female	Zone 6	EV ratio 6-8	13	BM1	0
18	Female	Zone 6	EV ratio 9-12	16	BM4	0
18	Female	Zone 6	EV ratio 13-15	13	BM2	0
18	Female	Zone 6	EV ratio 16-19	2	BM1	0
18	Male	Zone 1	EV ratio <5	2	BM2	0
18	Male	Zone 1	EV ratio <5	3	BM2	0
18	Male	Zone 1	EV ratio <5	5	BM1	0
18	Male	Zone 1	EV ratio <5	5	BM5	0
18	Male	Zone 1	EV ratio <5	6	BM1	0
18	Male	Zone 1	EV ratio <5	8	BM2	0
18	Male	Zone 1	EV ratio <5	10	BM1	0
18	Male	Zone 1	EV ratio <5	15	BM2	0
18	Male	Zone 1	EV ratio 6-8	7	BM2	0
18	Male	Zone 1	EV ratio 6-8	7	BM3	0
18	Male	Zone 1	EV ratio 6-8	9	BM1	0
18	Male	Zone 1	EV ratio 6-8	10	BM3	0
18	Male	Zone 1	EV ratio 6-8	11	BM1	0
18	Male	Zone 1	EV ratio 6-8	11	BM2	0
18	Male	Zone 1	EV ratio 6-8	11	BM3	0
18	Male	Zone 1	EV ratio 6-8	12	BM1	1

Figure 4.2: First rows of our dataset

4.2 Data Oversampling

After cleaning the dataset, we partitioned the raw dataset, allocating 80% for training purposes and 20% for testing. The left-hand bar chart in Figure 4.3 illustrates the distribution of the **ClaimMade** variable within the raw training data. It reveals a significant imbalance, with class 0 representing 98.94% of observations, and class 1 only comprising 1.06%. This imbalance can adversely affect the performance of machine learning algorithms [Hanafy and Ming, 2021], due to the scarcity of class 1 data for effective learning and prediction.

To address this imbalance issue, we applied the random oversampling method on the training dataset. We used function ROSE from a library called ROSE in R to balance the response data. According to ROSE, synthetic balanced samples are produced [Lunardon and Torelli, 2014].

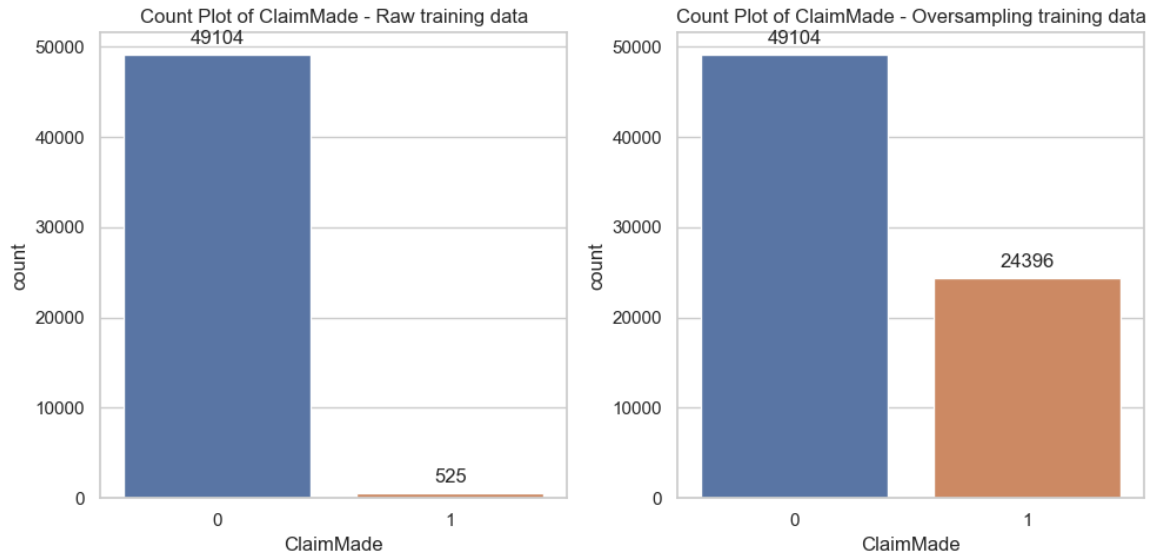


Figure 4.3: Proportion of ClaimMade for raw training data and oversampling training data

Balanced samples are generated by random oversampling of minority observations which is class 1 in **ClaimMade**. After applying oversampling method, the proportion of **ClaimMade** for new training dataset is showing on the right-hand side in Figure 4.3. There are 24396 class 1 observations in the new training dataset, which accounts for 33% of the training dataset.

Subsequently, we used the oversampling training dataset to train our machine learning models, reserving the test dataset for performance evaluation. Figure 4.4 visualises the distribution of **ClaimMade** in the test dataset, revealing a stark disproportion, with only 131 class 1 observations compared to 12,276 class 0 observations. Consequently, our model evaluation will primarily focus on assessing the accuracy of predicting class 1 observations.

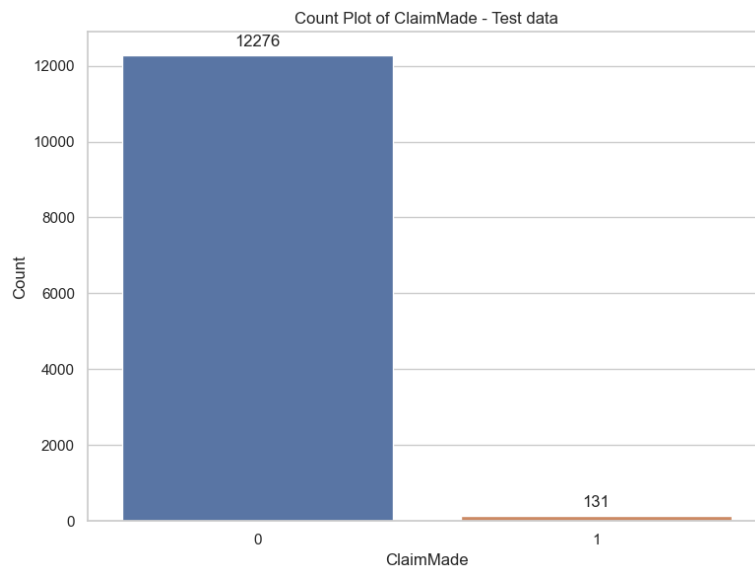


Figure 4.4: The proportion of ClaimMade for test data

4.3 Exploratory Data Analysis

We will start by describing the overall trends of the data before our analysis using XAI. This section will focus on the raw data, using both the train and test data together, rather than the oversampled data. Figure 4.5 illustrates the distribution of each variables in the dataset. We



Figure 4.5: The distribution of each variable in the dataset, raw data

can see from the graph of the average claim occurrence by **OwnerAge** that the general trend is that the average claim occurrence seems to decrease as the age increases. There seems to be a peak from ages 20-30 with the average claim occurrence decreasing after that. As we increase the age past 50 there seems to be bigger fluctuations within the data, which could suggest a lack of data, which we know is true. The average claim occurrence by **VehAge** is a more clear trend of a decrease in claim occurrence as the age of the vehicle increases. It is hard to discern any more than a linear relationship between the response and explanatory variable without further analysis. We can see that the average claim occurrence is higher for Male than for Female. Note also that most of our data comes from men; only 11% of the total exposure comes from women. We can see that the average claim occurrence peaks from EV ratios 16-19 and 20-24, in Figure 4.5. Besides, one can see that the average claim occurrence remains relatively constant for all Bonus Classes except for BM7 where it is shown in Figure 4.5, being significantly higher than the rest of the classes. There is a clear trend in the average claim occurrence by **Area**, peaking in Zone 1 and steadily decreasing as we increase the zone number - Zone 1 is the central areas of Sweden's three biggest cities, and decreases as the density of the areas decreases. We will be analysing this trend further in Section 4.5.2. One could claim that these plots may not be relevant since our analysis is based on the oversampled data but you can see from Figure 4.6 that the plots and trends are very similar trends in these plots.

We can now see in Figure 4.7 histograms of each of the variables, split by gender. We can instead see the count of the claims made in each category, rather than the average claim occurrence. There are many similarities but in this section we will note any interesting differences, between these histograms and the previous line and bar charts. Firstly, note the count for **OwnerAge** between 40 and 60 is much higher than the average claim occurrence for the same age range. This shows that the amount of data for this age range is particularly large but the claim rate remains quite low. We can see a similar trend in the **VehAge** graph, with many claims at age 15, but still a lower claim occurrence rate. This suggests that claims are more likely to

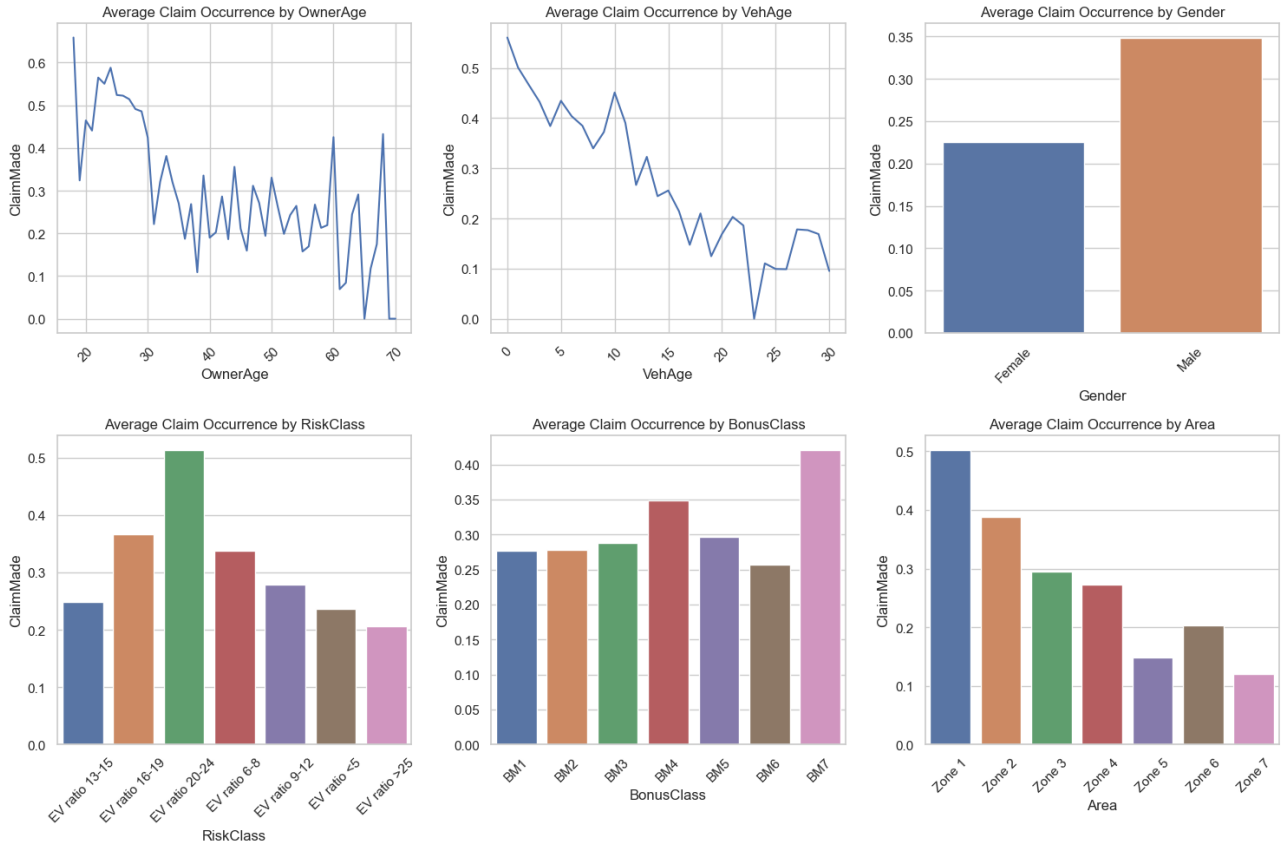


Figure 4.6: The distribution of each variable in the dataset, oversampled data

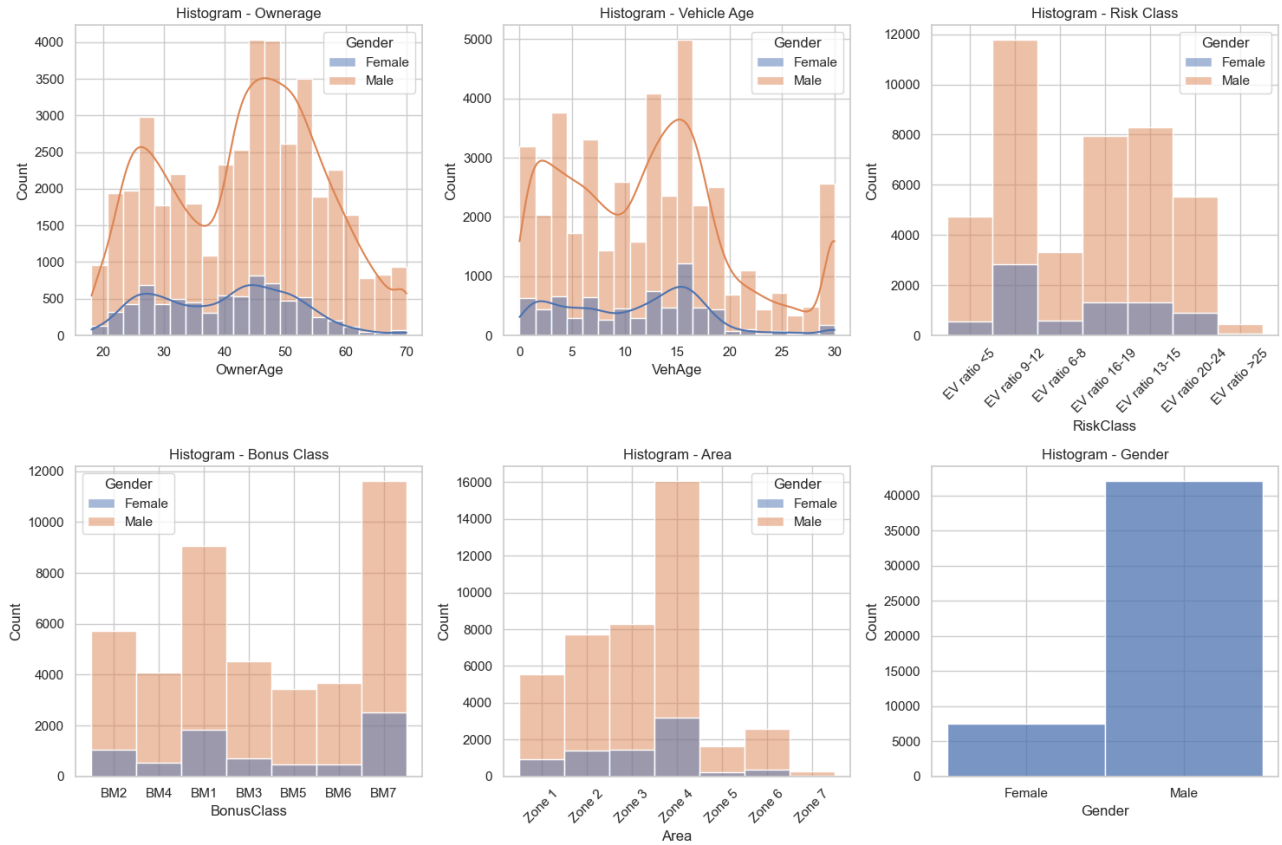


Figure 4.7: The distribution of each variable in the dataset, raw data, split by gender

occur with a lower vehicle age. Interestingly, Zone 4 in the **Area** graph has a very high count for the number of claims made but isn't outstanding in the average claim occurrence graph, it is the earlier zones that have the highest average claim occurrence. Finally, we can note that the number of claims for men is very high since as we know the data is mainly gathered from men. The average claim occurrence for women is not as far away from the men as Figure 4.7 would suggest.

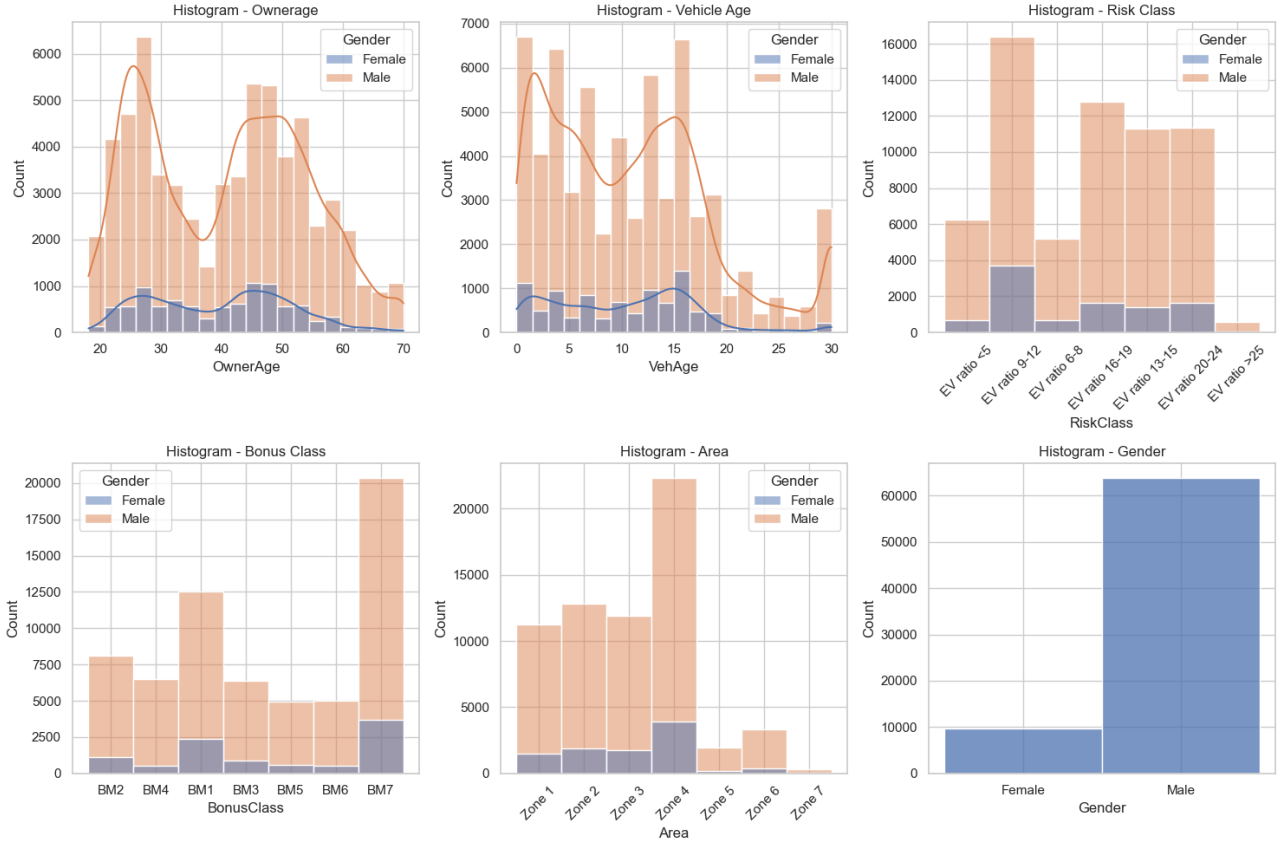


Figure 4.8: The distribution of each variable in the dataset, oversampled data, split by gender

There is an interesting observation to be gained by looking at Figure 4.8. The count is much higher for **OwnerAge** 20-30 than in the raw data. This is most likely because the oversampling technique is sampling data from here as this driver age range is the most likely to make a claim. We can see a similar trend in the **VehAge** graph, where the lowest vehicle age are upweighted when compared to the raw data.

4.4 Model Performance

In this project, we trained four machine learning classification models to predict the claim occurrence. We used the `scikit-learn` library in Python, all models were trained using default settings. In this section, we will assess the performance of these models across various dimensions including recall, precision, f1, accuracy and AUC metrics as well as confusion matrices and ROC curve. Then we compared all classifier models and made conclusions of which model is the best classifier.

Figure 4.9 displays the values of different metrics using test dataset for the four models. Examining the first two columns of the table, recall measures the effectiveness of the model in identifying all relevant instances of a particular class by evaluating the proportion of correctly classified instances of that class among all instances actually belonging to that class. On the other hand, precision evaluates the accuracy of the classification for a specific class by assessing

	recall	precision	f1	accuracy	auc
LogisticRegression	0.427481	0.033473	0.062084	0.863625	0.754558
RandomForestClassifier	0.274809	0.043849	0.075630	0.929072	0.752981
SVC	0.381679	0.026824	0.050125	0.847264	0.616956
GradientBoostingClassifier	0.458015	0.038437	0.070922	0.873297	0.764311

Figure 4.9: Model Performance, test data

the proportion of correctly classified instances among all instances predicted as belonging to that class. Mathematically, recall and precision have the following equations:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4.1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4.2)$$

	recall	precision	f1	accuracy	auc
LogisticRegression	0.467946	0.641384	0.541107	0.736558	0.773422
RandomForestClassifier	0.691384	0.848996	0.762126	0.856748	0.942913
SVC	0.784760	0.736941	0.760099	0.835578	0.822793
GradientBoostingClassifier	0.548082	0.698845	0.614349	0.771605	0.834548

Figure 4.10: Model Performance, train data

Comparing the values of recall and precision for the four models, it is worth noting that the precision values are much lower than the recall values. This is because the response variable in the test dataset is highly imbalanced, leading to the minority class (class 1) being underrepresented compared to class 0. In such cases, models may have a tendency to predict the majority class more frequently, resulting in a low precision value.

In terms of model comparison, the random forest classifier has the lowest recall value but the highest precision value among the four models. The gradient boosting classifier has the highest recall value and the second-highest precision. It is difficult to determine which model is better at predicting the True Positive value. Therefore, we will examine the F_1 score, which is shown in the third column of the model performance table. On the other hand, focusing on the smallest FN values, it is possible to continue with gradient boosting classifier. In all cases, regarding the imbalanced nature of testing data, no model is close to the ideal. For the further analysis, we mainly stick to the random forest classifier.

F_1 is another measure of model performance that combine both recall and precision, mathematically, it has the following calculation:

$$F_1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = \frac{2TP}{2TP + FP + FN} \quad (4.3)$$

According to the equation, F_1 is the harmonic mean of precision and recall. It is a commonly used metric to evaluate a model's predictive ability. From the table, we observe that the random forest classifier has the highest F_1 score compared to other models. It's worth noting that the

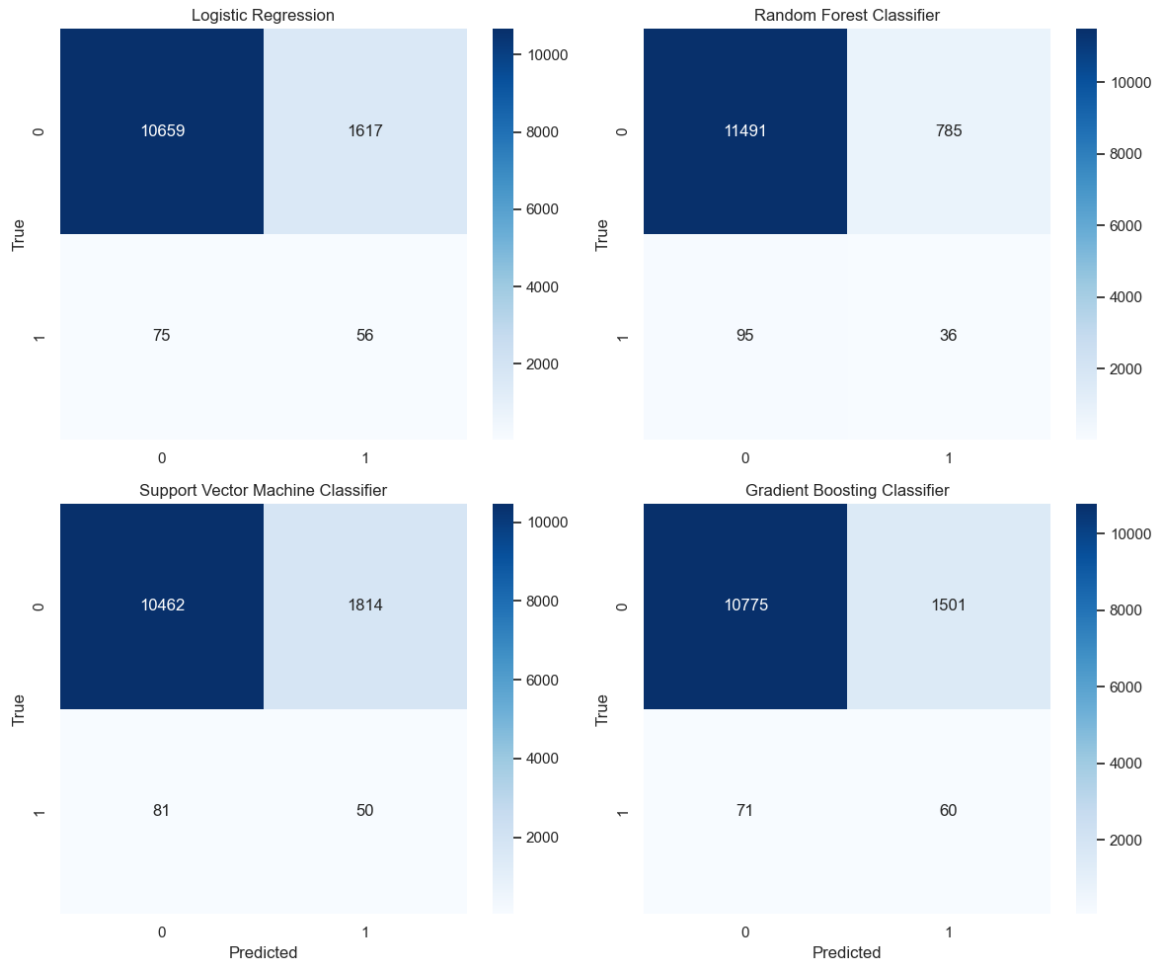


Figure 4.11: Confusion Matrix visualization (Top-left: Logistic, Top-right: RF, Bottom-left: SVC and Bottom-right: GBC)

gradient boosting classifier has the second-highest F_1 score, with a value of 0.070922, which is close to the value of the random forest classifier.

Comparing the values with Figure 4.10, which displays the metrics using the train dataset for the four models, we observe that SVC has the highest recall value, while the random forest classifier has the highest precision. In terms of F_1 score and accuracy, the random forest classifier still ranks first in this comparison. It should be noted that the precision and F_1 score are much greater than those values shown in Figure 4.9. This difference is because the response variable is balanced to a certain extent compared to the test dataset.

Moreover, to further understand the predictive ability of the four models, we delve into the confusion matrix shown in Figure 4.11. Firstly, the random forest classifier has made the most correct predictions for class 0, totaling up to 11,491 cases, significantly higher than the predictions made by the other models. This indicates that the random forest classifier is the best model for predicting class 0. However, this model has the least number of correct predictions in terms of class 1, with only 36 cases. In contrast, the gradient boosting classifier made 60 predictions for class 1 when the actual class is also 1. Therefore, we can conclude that the random forest classifier performs well in correctly identifying instances of the majority class (class 0) when they are actually class 0, while the gradient boosting model performs well in correctly identifying instances of the minority class (class 1) when they are actually class 1.

Accuracy in Figure 4.9 measures how many times a model made a correct prediction across the test dataset. However, this metric can only be reliable when the dataset is balanced, meaning the response variable **ClaimMade** must be balanced for this to be reliable.

According to the table, the random forest classifier has the highest accuracy with a value of 0.929. This indicates that the random forest classifier made the most correct predictions even

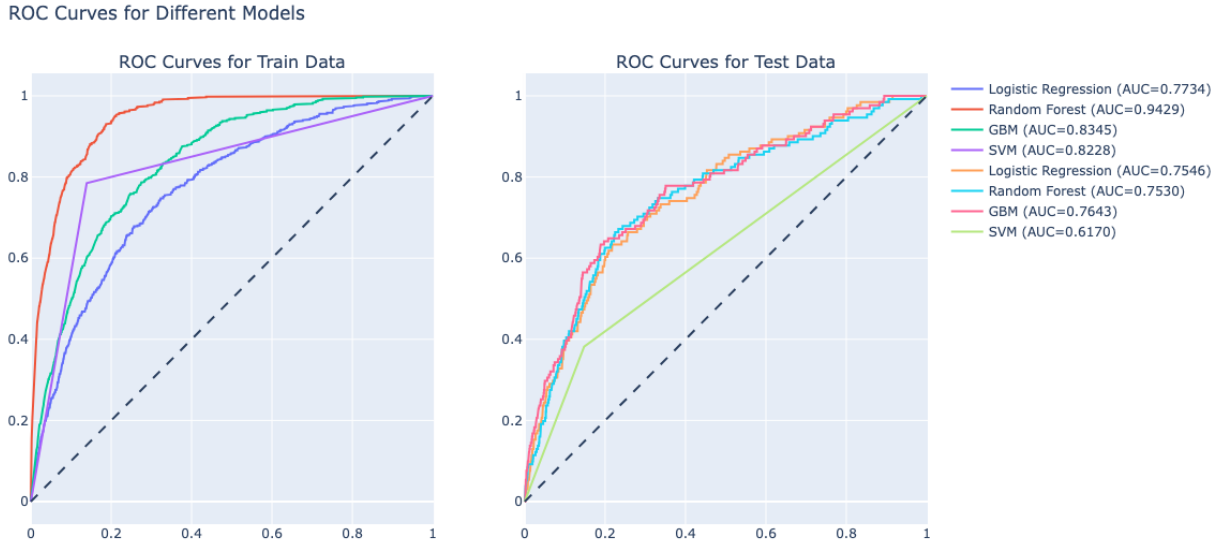


Figure 4.12: ROC curve comparison of models for train (left-panel) and test data (right-panel)

though the test dataset is imbalanced.

Let's examine the AUC value and ROC curve shown in Figure 4.12. The ROC curve, representing the Receiver Operating Characteristic curve, illustrates the True Positive Rate versus False Positive Rate, while AUC measures the area under this curve. A higher AUC value indicates better predictive performance, specifically in correctly identifying class 0 as 0 and class 1 as 1.

By analysing the model performance table and ROC curve from Figure 4.10 and the left-panel in Figure 4.12, we observe that random forest classifier has the highest AUC value compare to other models. While looking at the model performance on test data from Figure 4.9 and right-panel in Figure 4.12, we observe that both logistic regression and random forest classifier demonstrate nearly identical AUC values, while the gradient boosting classifier achieves the highest AUC with a value of 0.764. It's noteworthy that the SVC model shows the lowest AUC in this comparison.

In conclusion, based on the comparison of different metrics, the random forest classifier emerges as the optimal model for predicting claim occurrence in terms of precision or f1 score. However, despite its high F_1 score, accuracy, and one of the highest AUC values, it shows limitations in predicting the minority class, as evident from the confusion matrix. On the other hand, if recall is the main metric, one can prefer gradient boosting classifier. Therefore, for further analysis, we will use random forest classifier for further analysis in the subsequent sections, with further analysis using the gradient boosting classifier shown in the appendix.

4.5 Explainability

In this section, we will start using DALEX package to explore and explain the models and evaluate how the models, especially the random forest classifier, perform at the instance-level and dataset-level.

4.5.1 Instance-Level Exploration

Instance-level exploration methods help us understand how a model yields a prediction for a particular single observation [Biecek and Burzykowski, 2021]. In this section, we will analyse

how the model’s prediction for a particular instance differs from the average prediction and how the difference can be distributed among predictor variables. Moreover, we can investigate how the model’s prediction changes if the value of a single predictor variable changes.

To investigate how each variable makes contributions to the prediction, we employed Break-Down plots to illustrate the effects of the variables. The main idea of break-down plots is to generate order-specific explanation of explanatory variables contribution. It is important to consider the ordering for these reasons: In non-additive models, the order of features in explanations plays a crucial role. The comprehension of model reasoning relies heavily on the order in which explanations are presented. Establishing a suitable order can enhance the clarity and comprehension of predictions. Typically, human perception focuses on only a handful of variables when considering predictions. Hence, it’s vital to emphasize the most significant features and relegate less important variables to the end of explanations Gosiewska and Biecek [2019].

This instance, shown in Figure 4.13, has a final prediction with 0.172 probability to have a claim by random forest classifier, which indicates that the selected instance has no claim and the model successfully made a correct prediction. Now, we would like to understand which explanatory variable drove this prediction and the effects of different orderings. Two potential explanations are given below.

OwnerAge	Gender	Area	RiskClass	VehAge	BonusClass
26	Female	Zone 4	EV ratio 9-12	2	BM7

Figure 4.13: Selected instance variable

Explanation 1: We first consider the explanatory variables **OwnerAge**, **VehAge**, **Gender**, **Area**, **BonusClass**, and **RiskClass**, in this order. Figure 4.14 indicates positive contributions for the first two variables and negative contributions for **Gender** and **Area**. Thus, the fact that the owner is aged 26 with a motorcycle age of 2 increases the probability of having a claim. However, being female and living in zone 4 substantially lowers the probability of a claim occurring. Being classified as BM7 in terms of **BonusClass** and with an EV ratio of 9-12 for the **RiskClass** both slightly increase the probability.

This last conclusion is the result of the fact that younger age with less experience in motorcycle driving tends to have a higher likelihood of making a claim. However, being female, younger, and living in zone 4 lowers the chances of having a claim. This may be because younger females tend to be more cautious than males of the same age.

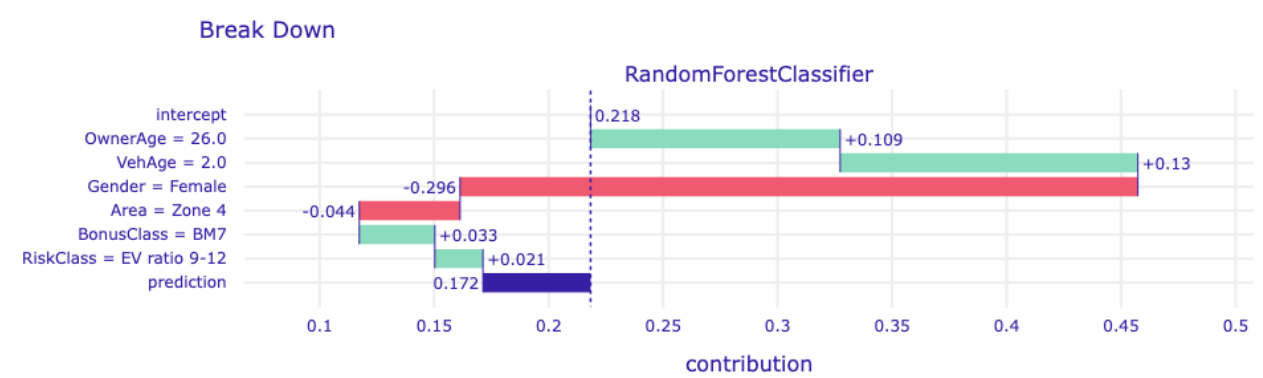


Figure 4.14: Break-down Plot with order 1

Explanation 2: We now consider the following order of explanatory variables: **Area**, **Gender**, **OwnerAge**, **VehAge**, **BonusClass**, and **RiskClass**. Figure 4.15 indicates a positive contribution of **Area**, **Gender**, and **OwnerAge**, unlike in the first explanation. Again, the fact that the owner is female living in zone 4 area decreases the chances of having a claim. However, given the owner’s gender is female, a younger owner age decreases the probability of claim occurrence. Finally, **VehAge** at 2, **BonusClass** at BM7, and **RiskClass** at EV ratio of 9-12 further increase the probability of a claim; the contributions are much higher than in the first explanation.

This last conclusion stems from the fact that a female with a younger age has a very low probability of having a claim, while having a lower motorcycle age increases the chances of a prediction.

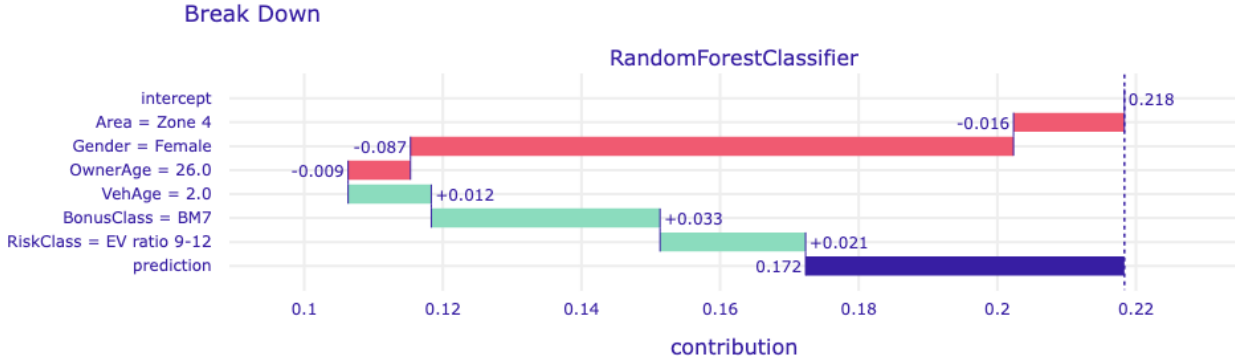


Figure 4.15: Break-down Plot with order 2

Break-down Plots for Interaction

In the previous section, we presented two break-down plots with different orders and discussed the attributions of different explanatory variables to the random forest classifier’s predictions. However, the results of the method introduced above depend on the ordering of the explanatory variables used in computations. Moreover, there might be potential interactions between different explanatory variables that should not be neglected. Therefore, we introduce an algorithm to identify interactions between pairs of variables and take them into account when constructing break-down plots [Biecek and Burzykowski, 2021]. We will analyse whether there are any interactions between owner information and motorcycle information. Specifically, we want to investigate whether there are any interactions between **OwnerAge** and **Gender** or **RiskClass**, **VehAge**.

Firstly, let’s look at the net contribution of each variable without any interaction terms at Figure 4.16

Then, Figure 4.17 shows the calculation results of the potential interaction variables for this selected instance. Base on the ordering of the rows, the following sequence of variables is identified as informative:

1. **Gender:OwnerAge** Interaction: This interaction variable pertains to the owner’s personal information and has the largest (in absolute value) net contribution, equal to 0.095.
2. **VehAge:RiskClass** Interaction: This interaction term relates to motorcycle information and is the second largest (in absolute value) net contribution, equal to 0.069.
3. **BonusClass**: The net contribution (in absolute value) for this variable is 0.038.
4. **Area**: This variable has a net contribution (in absolute value) of 0.058 and is placed in the last position.

	variable_name	variable_value	variable	cumulative	contribution	sign	position	label
0	intercept		intercept	0.218331	0.218331	1.0	7	RandomForestClassifier
1	OwnerAge	26.0	OwnerAge = 26.0	0.327403	0.109072	1.0	6	RandomForestClassifier
2	Gender	Female	Gender = Female	0.122984	-0.204419	-1.0	5	RandomForestClassifier
3	VehAge	2.0	VehAge = 2.0	0.161890	0.038906	1.0	4	RandomForestClassifier
4	BonusClass	BM7	BonusClass = BM7	0.198854	0.036964	1.0	3	RandomForestClassifier
5	RiskClass	EV ratio 9-12	RiskClass = EV ratio 9-12	0.229903	0.031049	1.0	2	RandomForestClassifier
6	Area	Zone 4	Area = Zone 4	0.171961	-0.057942	-1.0	1	RandomForestClassifier
7			prediction	0.171961	0.171961	1.0	0	RandomForestClassifier

Figure 4.16: Without Interaction Results for Random Forest classifier

	variable_name	variable_value	variable	cumulative	contribution	sign	position	label
0	intercept		intercept	0.218331	0.218331	1.0	5	RandomForestClassifier
1	Gender:OwnerAge	Female:26.0	Gender:OwnerAge = Female:26.0	0.122984	-0.095346	-1.0	4	RandomForestClassifier
2	VehAge:RiskClass	2.0:EV ratio 9-12	VehAge:RiskClass = 2.0:EV ratio 9-12	0.192183	0.069198	1.0	3	RandomForestClassifier
3	BonusClass	BM7	BonusClass = BM7	0.229903	0.037720	1.0	2	RandomForestClassifier
4	Area	Zone 4	Area = Zone 4	0.171961	-0.057942	-1.0	1	RandomForestClassifier
5			prediction	0.171961	0.171961	1.0	0	RandomForestClassifier

Figure 4.17: Interaction Results for Random Forest classifier

Figure 4.18 displays the break-down plot with the interaction variables and their contributions to the model’s prediction, corresponding to the results shown in Figure 4.17. The interaction between **Gender** and **OwnerAge**, and **VehAge** and **RiskClass** variables are depicted in the plot as single bars.

Figure 4.16 illustrates that **OwnerAge** alone would increase the mean prediction by 0.109, while **Gender** would decrease the mean prediction by 0.204. When considered together, they decrease the average prediction by approximately 0.095, which aligns with the net contribution of the interaction term.

Similarly, the single net contribution of **VehAge** is 0.0389, and 0.0310 for **RiskClass**, resulting in an accumulated net contribution of approximately 0.0699. This value is nearly identical to the net contribution of the interaction term.

Hence, we can conclude that there are no interactions between these explanatory variables for this selected instance, as their single net contributions equal the interaction terms.

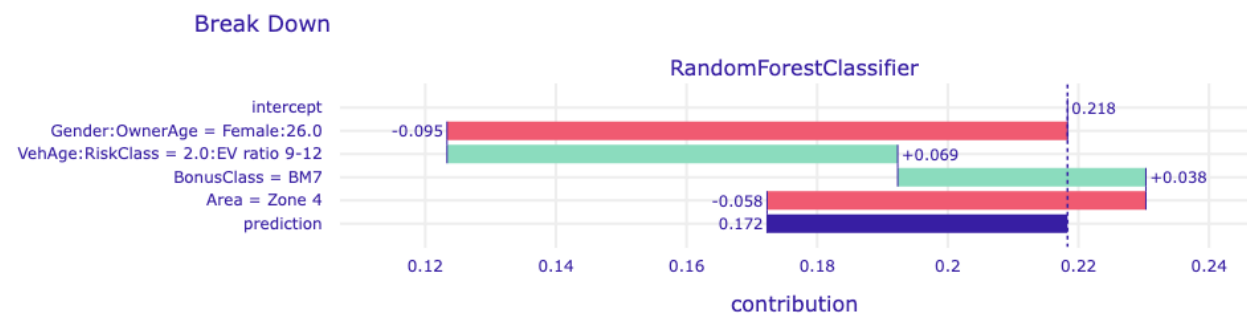


Figure 4.18: Break-down Plot for Interaction

Shapley Value

In the previous chapters, we introduced break-down plots with different orderings and interaction terms. However, the computed attribution values depend on the order of explanatory variables used in the calculations. One solution to this issue is to find an ordering in which the most important variables are placed at the beginning. To further understand how each explanatory variable contributes to the model’s prediction, we will introduce another approach; Shapley values.

Shapley values are one of the most commonly used techniques for evaluating a black-box machine learning model [Mayer et al., 2023]. This method involves averaging the value of a variable’s contribution over all possible orderings [Biecek and Burzykowski, 2021]. In this section, we will analyse the same instance. Figure 4.19 displays the calculation results and the averaging process to obtain the Shapley values. By default, Shapley values are calculated and plotted for all variables in the data.

	variable	contribution	variable_name	variable_value	sign		label	B
0	VehAge = 2.0	0.082406	VehAge	2	1.0	RandomForestClassifier	1	
1	BonusClass = BM7	0.101751	BonusClass	BM7	1.0	RandomForestClassifier	1	
2	Area = Zone 4	-0.012986	Area	Zone 4	-1.0	RandomForestClassifier	1	
3	OwnerAge = 26.0	0.097527	OwnerAge	26	1.0	RandomForestClassifier	1	
4	Gender = Male	0.059874	Gender	Male	1.0	RandomForestClassifier	1	
...	...	...	...	...	...	...	...	...
1	OwnerAge = 26.0	0.100037	OwnerAge	26	1.0	RandomForestClassifier	0	
2	BonusClass = BM7	0.096651	BonusClass	BM7	1.0	RandomForestClassifier	0	
3	Gender = Male	0.036763	Gender	Male	1.0	RandomForestClassifier	0	
4	Area = Zone 4	-0.011467	Area	Zone 4	-1.0	RandomForestClassifier	0	
5	RiskClass = EV ratio 9-12	0.002883	RiskClass	EV ratio 9-12	1.0	RandomForestClassifier	0	

Figure 4.19: Shapley Values Results

Figure 4.20 shows the visualisation of the obtained Shapley values. From the plot, we can observe that, the **Gender** being female has the largest negative effect on the model’s prediction, which is up to 0.198. While the explanatory variables **VehAge** at 2, **BonusClass** at BM7 and **OwnerAge** at 26 both have positive contributions. Then, the last two explanatory variables **Area** and **RiskClass** have the negative contribution with the lowest values. It is worth noting that, **OwnerAge** has a negative contribution in Figure 4.14, but has a positive contribution of 0.034 showing in the Shapley value method.

Connecting to the real-life scenario, Figure 4.20 shows that females are less likely to have a claim than males. The probability of having a claim increases with less motorcycle driving experience (**VehAge** is low) and when the owner is young. Moreover, living in Zone 4 areas, which are small towns and countryside, also lowers the probability of having a claim due to the lower traffic pressure in these areas.

Therefore, analysing instances via Shapley values provides a thorough approach to decomposing a model’s prediction into the contributions of different explanatory variables which are additive.

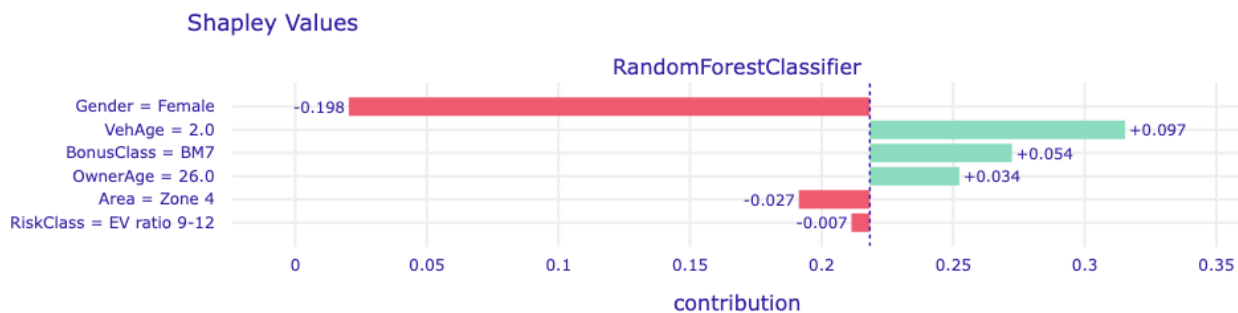


Figure 4.20: Shapley Values Plot for Random Forest classifier

Comparison with different explanatory variables

We have analysed the Shapley values for the selected instance, as shown in Figure 4.20. Now, we are interested in understanding the effects of changing the value of a single explanatory variable. For example, what are the Shapley values for each explanatory variable if the **Gender** is male? In this section, we will compare the selected instance with three other instances where we change one of these variables: **Gender**, **OwnerAge**, and **VehAge** one at a time.

Figure 4.21 displays the four sets of Shapley values, with the top one representing the original instance. Let's first examine the second instance, where we changed the **Gender** to male. Here, we observe that the top three explanatory variables maintain the same positive contribution as the selected instance. However, in this case, **Gender** contributes positively to the model's prediction. This suggests that, under all other conditions being equal, being male increases the probability of having a claim.

Next, we change the **OwnerAge** to 40 and compare it to the original instance where **OwnerAge** is 26. The third Shapley value indicates that **OwnerAge** now has a negative contribution, implying that an older female is less likely to make a claim compared to a younger female if assuming all other conditions are the same.

Lastly, we change the **VehAge** to 18, contrasting with the original instance where **VehAge** is 2. Unlike the largest positive contribution of **VehAge** at 2, having a **VehAge** of 18 significantly decreases the probability of having a claim, resulting in a negative contribution with a value of 0.078. This could be because a higher **VehAge** implies more driving experience, which might reduces the likelihood of making mistakes that could lead to damage or collisions on the road. As a result, the chance of getting a claim decreases.

In conclusion, these comparisons highlight how individual explanatory variables can impact the model's predictions. Understanding these effects can provide valuable insights into the factors influencing claim occurrence and inform decision-making processes in insurance risk assessment. However, it is crucial to note that the considered model is subject to the criticism of having poor testing data performance.

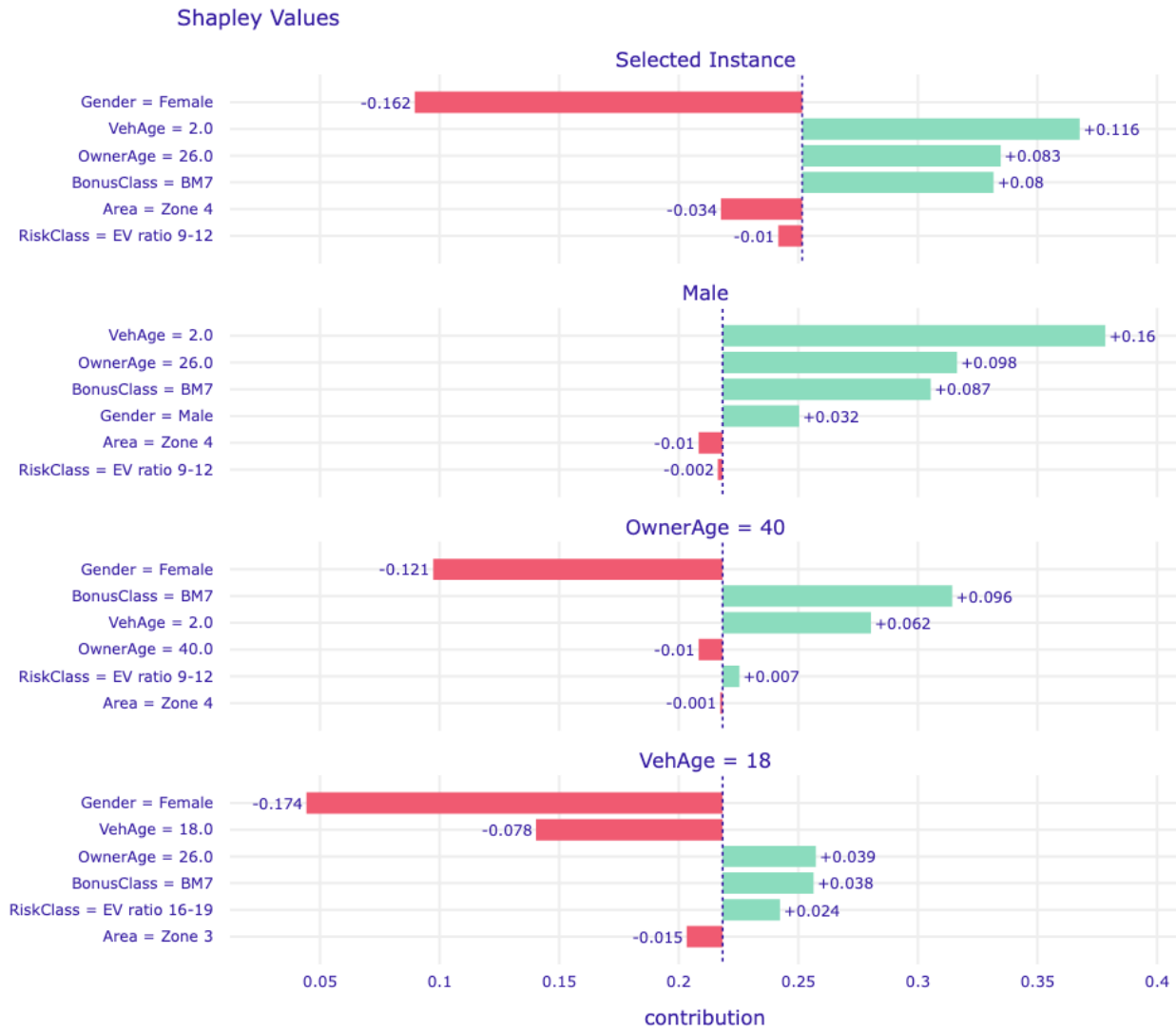


Figure 4.21: Shapley value with comparison

Ceteris-paribus Profiles

After analysing the effects of changing a single explanatory variable on the model's prediction using Shapley values, we now turn our attention to understanding the effects of different values of the explanatory variables. This exploration is facilitated by a method known as ceteris paribus profiles (CP profiles), which examines the influence of an explanatory variable while assuming that the values of all other variables remain constant [Biecek and Burzykowski, 2021]. The objective is to discern how changes in the values of a variable impact the model's prediction.

In Figure 4.22, we present the CP profiles for the **OwnerAge** and **VehAge** variables for the random forest classifier. Each curve in the plot illustrates how the model's prediction would change as the value of a specific explanatory variable varies, with the instance of interest represented by a dot on each curve. This instance corresponds to the selected case analysed in the previous section.

By observation, the profile for the random forest classifier demonstrates a step function with some variability. The plot clearly illustrates the trend, indicating that the instance has a prediction of 0.17, as denoted on the y-axis by the dot. For instance, if all other explanatory variables remain unchanged, a female aged 45 would have a predicted probability of approximately 0.27, still lower than 0.5, implying no claim occurrence. Conversely, if the motorcycle's age is 20, the prediction drops significantly to around 0.075.

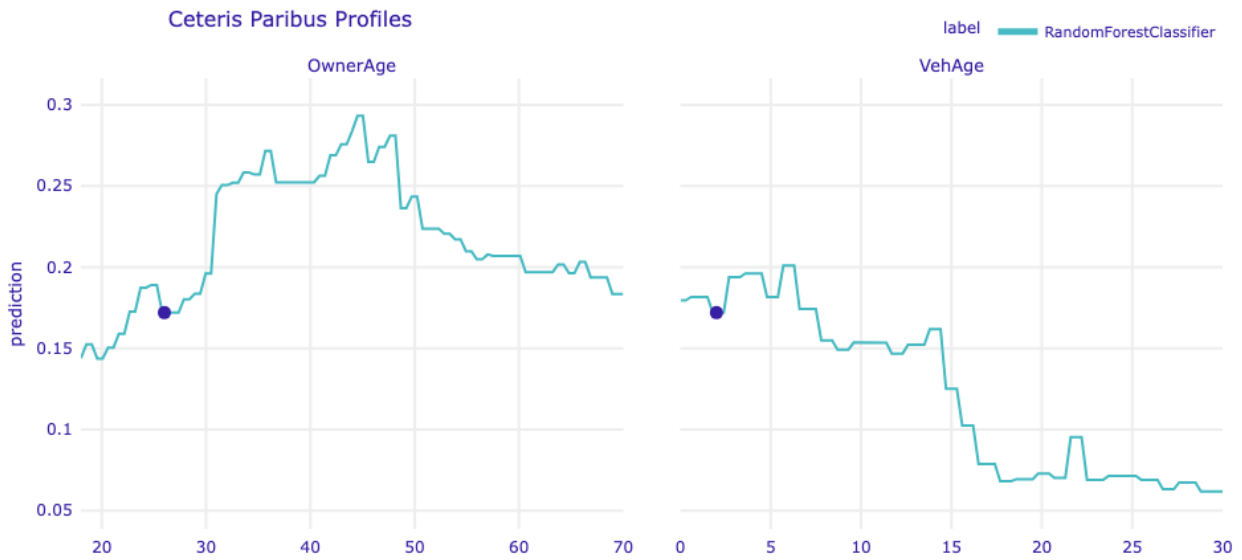


Figure 4.22: CP profiles for random forest classifier. Left panel for **OwnerAge**, right panel for **VehAge**

Figure 4.23 displays the CP profiles for all four models, not just random forest. It is noteworthy that the profile for the logistic regression model appears as a smooth line, while that for the SVC model is a straight line with all predictions at 0. In contrast, the CP profiles for the random forest classifier and gradient boosting classifier are both step functions with greater variability than the other two models. Although the gradient boosting classifier has a higher average prediction than the random forest model, the general shape of their CP profiles is similar. All models predict a probability lower than 0.5 for this instance, effectively indicating the absence of a claim.

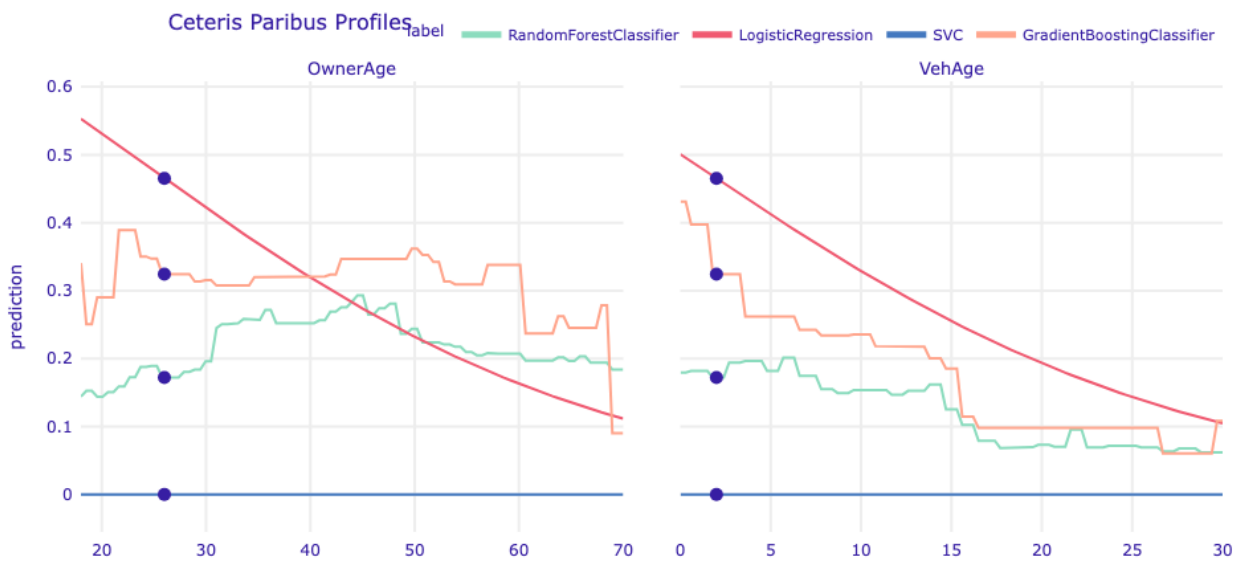


Figure 4.23: CP profiles for all models. Left panel for **OwnerAge**, right panel for **VehAge**

After exploring the effects of numerical explanatory variables through CP profiles, we now shift our focus to understanding the impacts of the categorical variables. Figure 4.24 presents the CP profiles for the four categorical variables, utilising bar plots to depict their influences.

Beginning with **Area**, we observe significant variations in the predicted probability for the random forest classifier when the value of **Area** changes. Specifically, transitioning to ‘Zone 1’ and ‘Zone 2’ results in a notable increase in the predicted probability, while selecting ‘Zone 5’ leads to a decrease. Next, considering **BonusClass**, we find that all values except the current one result in a reduction in the predicted probability. Regarding **Gender**, a notable observation emerges: if this instance were male, the predicted probability would substantially increase, potentially exceeding 0.5. This underscores the considerable impact of **Gender** on the model’s prediction. Finally, for **RiskClass**, only the ‘EV ratio 20-24’ value leads to an increase in the predicted probability, while selecting other values results in a decrease.

In summary, by examining CP profiles, we gain insights into the influences of both numerical and categorical explanatory variables on the model’s prediction. This analysis enables a deeper understanding of the effects of each value of the explanatory variable.

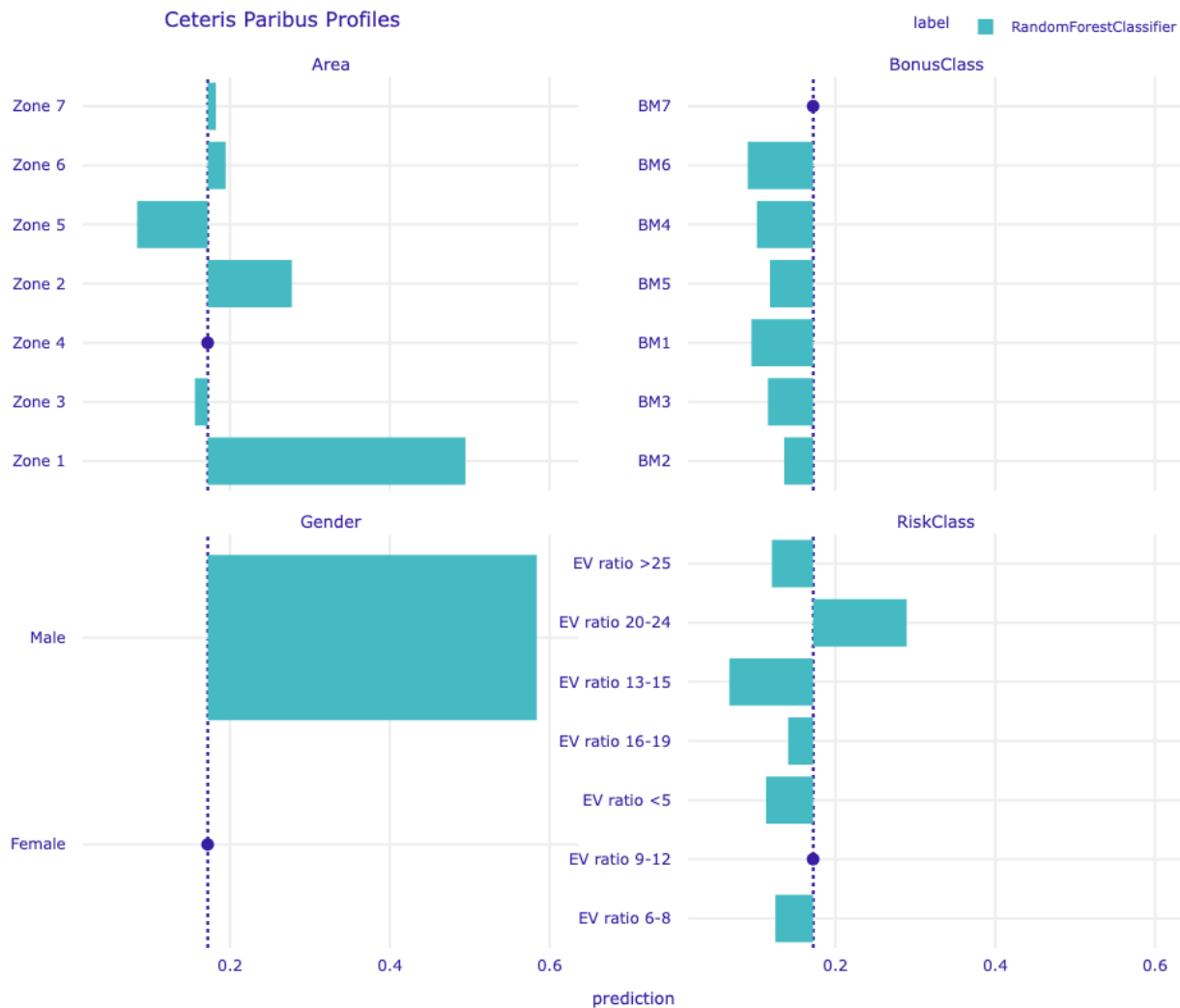


Figure 4.24: CP profiles for categorical variables

4.5.2 Dataset-Level

We will now analyse the dataset as a whole rather than focusing on one particular instance.

Variable Importance

We will start by analysing the overall variable importance. Figure 4.25 was calculated using the dropout loss function described in chapter 3. The plots suggest that the most important variable is **OwnerAge**. With the exception of the gradient boosting classifier, the next most important variable is **VehAge**. Surprisingly, **RiskClass** seems to be one of the less important variables which is interesting since the motorcycles are grouped by how ‘risky’ they are in terms of ‘likelihood’ to make a claim (actually by a calculation in relation to engine power and vehicle weight). At least by this measure, this is shown to not be the case. The gradient boosting classifier and SVC suggest **OwnerAge** is the most important variable by far while the logistic regression and random forest classifier suggest the importance is more spread out over the other variables like **VehAge** and even **Area**. We can note that in the random forest classifier, if variables are correlated, their effect is likely spread over all the correlated variables, whereas in the regularized-regression classifier like gradient boosting classifier, the most important variable is likely to dominate. This is most likely what has happened here with **OwnerAge** and **VehAge**. **Gender**, **BonusClass** and **RiskClass** are consistently the least important variables

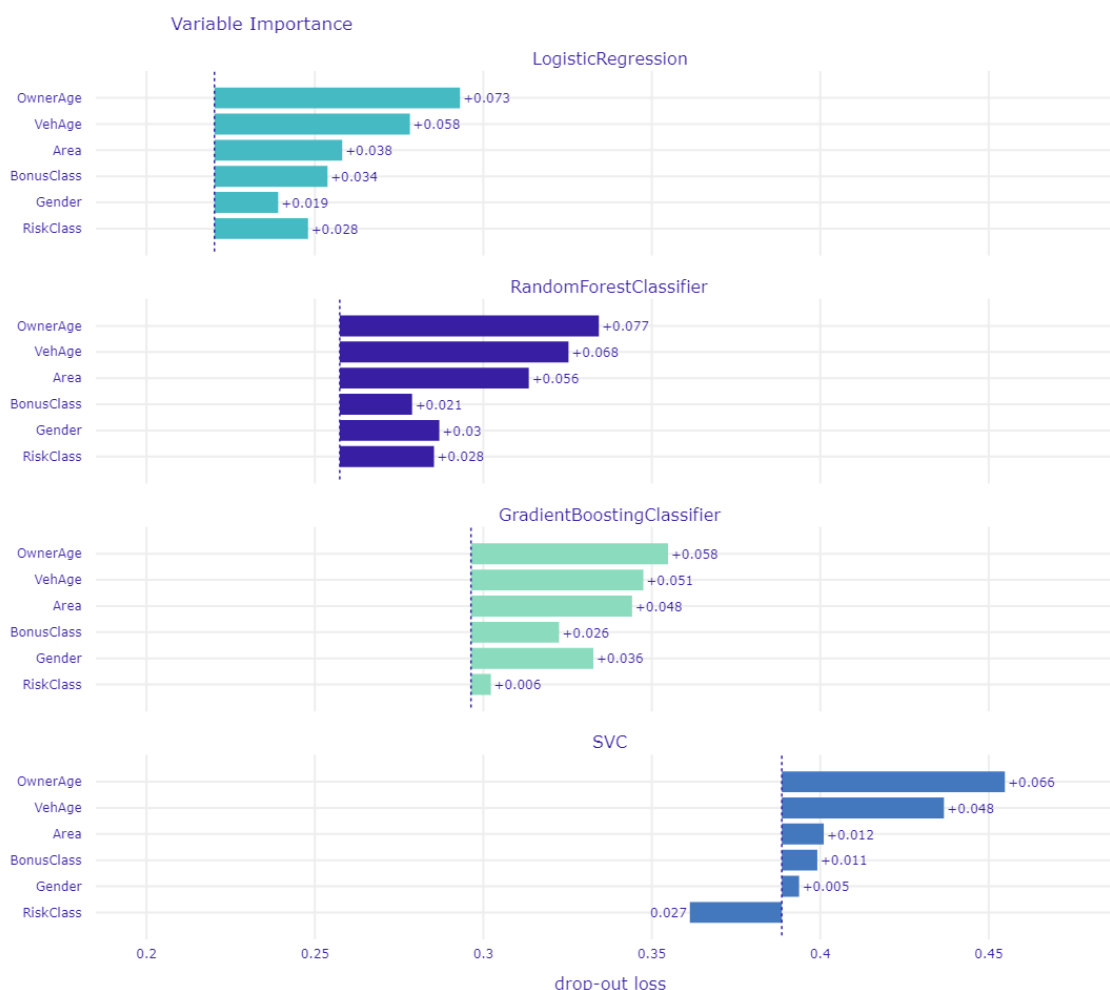


Figure 4.25: Variable importance plot via dropout loss function

Variable importance plots have a number of advantages; for example, they are very easy to understand and interpret. They are also relatively easy to calculate and so they can give a quick overview of what variables seem to be affecting the model's prediction and we can show all results from all the models on just one graph. The choice of loss function will also have an effect on what these graphs look like, we have chosen the dropout loss function but there are many other suitable loss functions which may produce differing results. One disadvantage of this method is the problem of permutations. With a different ordering of variables, this graph would look slightly different. We have averaged the effect of each variable over 10 orderings in this project, but even more accurate results could be obtained with increasing number of permutations.

Partial-dependence Profiles

Figure 4.26 shows the Partial Dependence profiles for all of the categorical variables in our dataset. One interesting result is that the **Gender** has a big disparity; men have a much higher prediction than the women in our dataset in general. This is something we can keep in mind during our fairness analysis. We must also note that most of the data in this dataset comes from the male gender.

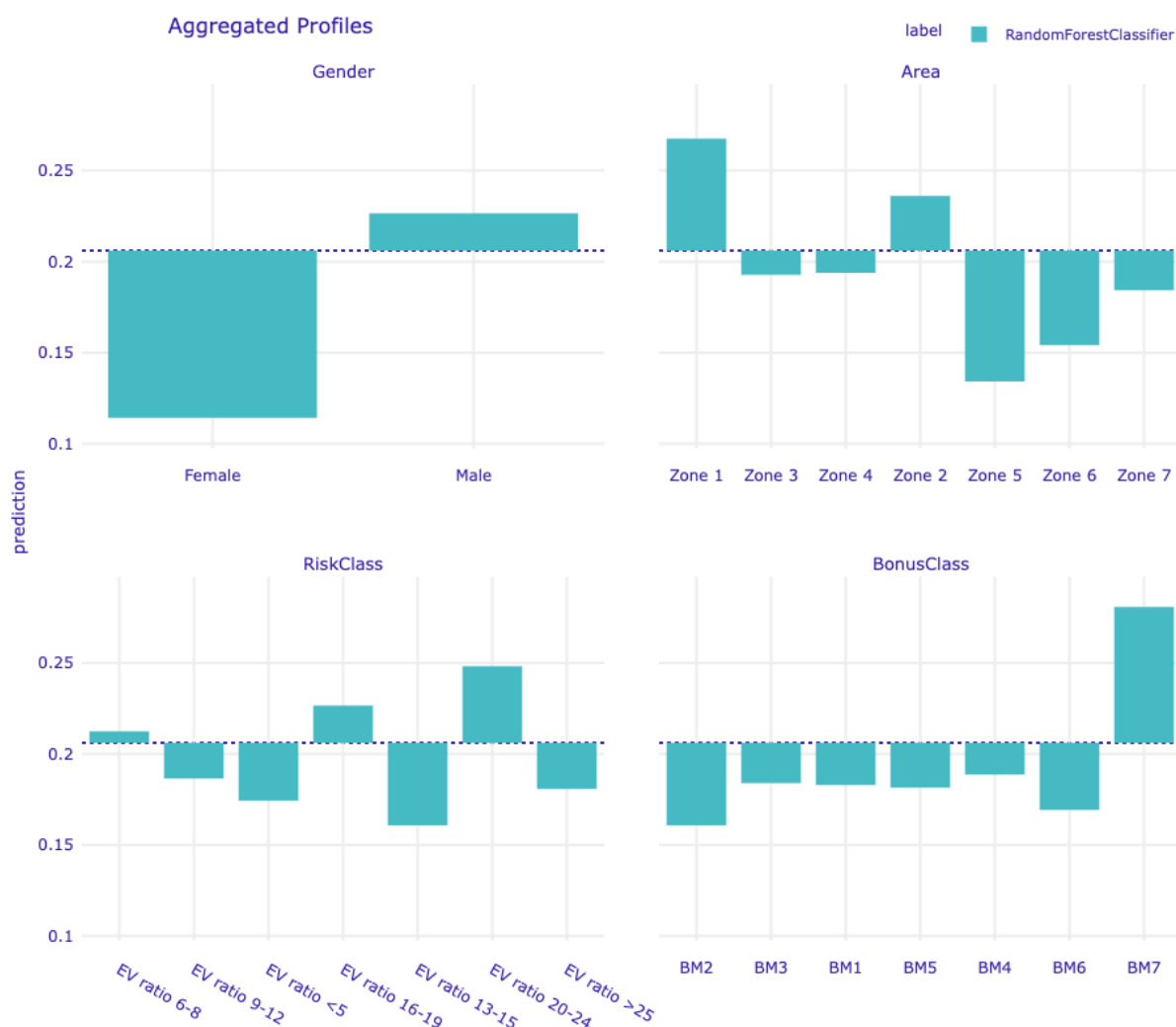


Figure 4.26: Partial Dependence plots for the categorical data, random forest model. Left panel for **OwnerAge**, right panel for **VehAge**

Another interesting result comes in the trend seen in the **Area** data. We can see that the more dense areas tend to have higher insurance claim predictions. Specifically, the mean prediction for this model is 0.223 and the prediction for Zone 1 is 0.283 and for Zone 2 it is 0.254. This makes sense as one is probably more likely to get into an accident in a busy city than in Sweden's more rural areas. Zone 1 is clearly the highest prediction and it the data obtained from Sweden's busiest cities. There does not seem to be much trend in the data from the **RiskClass** PD profile and given that this variable was often not very important in Figure 4.25, we will stop our analysis of this variable here. Finally, we can see that BM7 has the highest prediction rate at 0.280, which is very surprisingly as supposedly, BM7 is the most trusted motorcycle drivers who have evaded accidents over many years of having an insurance contract. All other bonus classes are below the mean prediction for this model. [Wüthrich and Merz, 2023] is unclear on exactly what the **BonusClass** represents and how it is calculated so it is hard to draw real-world conclusions about this category of the data.

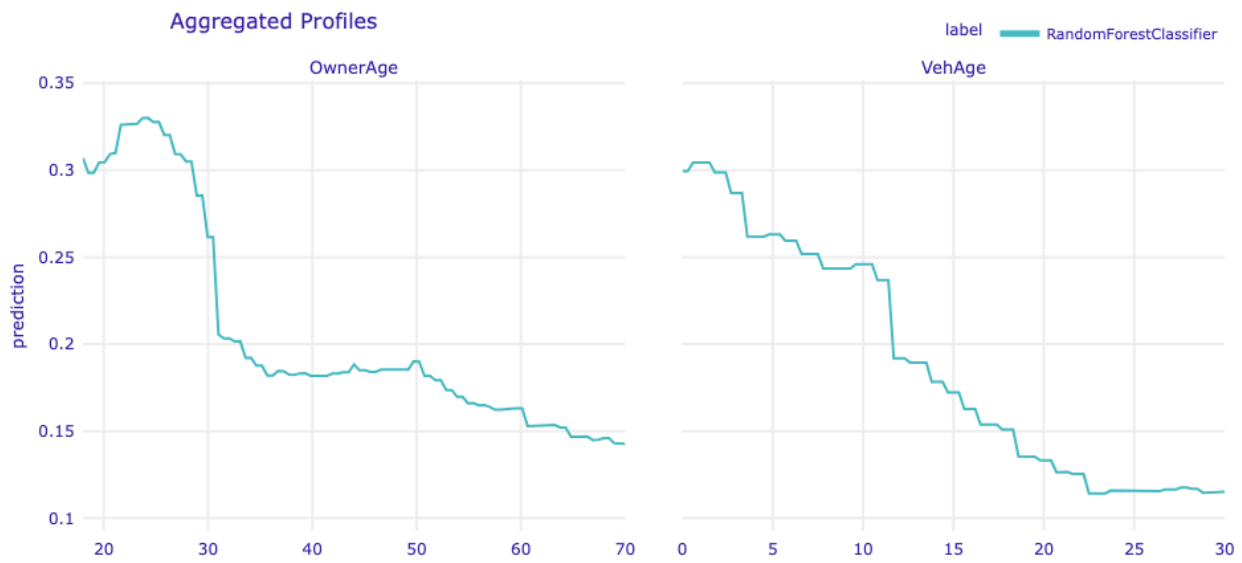


Figure 4.27: Partial Dependence plots for the numerical data, random forest classifier. Left panel for **OwnerAge**, right panel for **VehAge**

We can see the PD profiles for the numerical data in our data set, i.e. **OwnerAge** and **VehAge**. Figure 4.27 shows the average of 300 randomly selected *ceteris paribus* profiles of instances in our dataset. We can see the average trend decreasing as we had in section 4.3 with a lot less sensitivity to noise. In section 4.3 there were many fluctuations as the age of the owner increased, but this is not present here. We can see the data peaks at **OwnerAge** 25 with the prediction being 0.366 compared to the mean of 0.220. The peak of the data in the **VehAge** PD profile comes at the very start of the vehicle's lifecycle, at age 1, with a prediction of 0.323 compared to the same mean prediction. This makes sense as drivers who have more experience with their vehicle and on the road can navigate tough situations with more experience. Figure 4.27 shows a sharp dropoff in prediction around **OwnerAge** 30 and then consistent prediction after that. This is interesting as it suggests that the highest probability of a claim being made is when the driver is in their 20s. This would make sense as they would have less experience and would perhaps tend to be more reckless.

One of the categorical explanatory variables is **Area**. We want to investigate if the effect of **OwnerAge** and **VehAge** remains the same for each **Area**. Figure 4.28 shows the Partial Dependence profile for each of the 7 areas in the dataset for the random forest classifier. We can see they follow a similar pattern with peaks in the lower end of **OwnerAge** and **VehAge** with

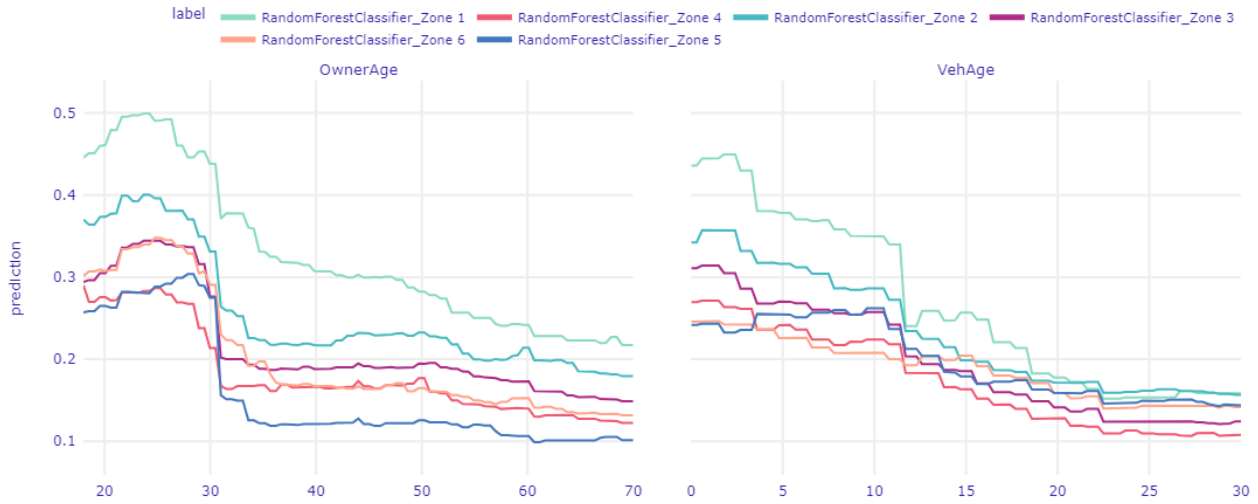


Figure 4.28: Partial Dependence profiles for the numerical data, grouped by the **Area**, random forest classifier. Left panel for **OwnerAge**, right panel for **VehAge**

dropoffs as the age increases. The most interesting conclusion to draw from this plot however, is that the predictions seems to be higher overall for the more dense areas, like Zones 1, 2 and 3, which represent Sweden's most busy areas, while Zones 5, 6 and 7 are consistently lower predictions. This agrees with Figure 4.26. Looking at the trends of each zone, they seem to follow the same pattern except perhaps Zone 1 in the **VehAge** plot. We could say that the peak is much higher for lower **VehAge** but then drops to an almost equal prediction for most vehicle ages above 15. This suggests that the **Area** does not have much of an effect on the prediction of **VehAge** if the age is large.

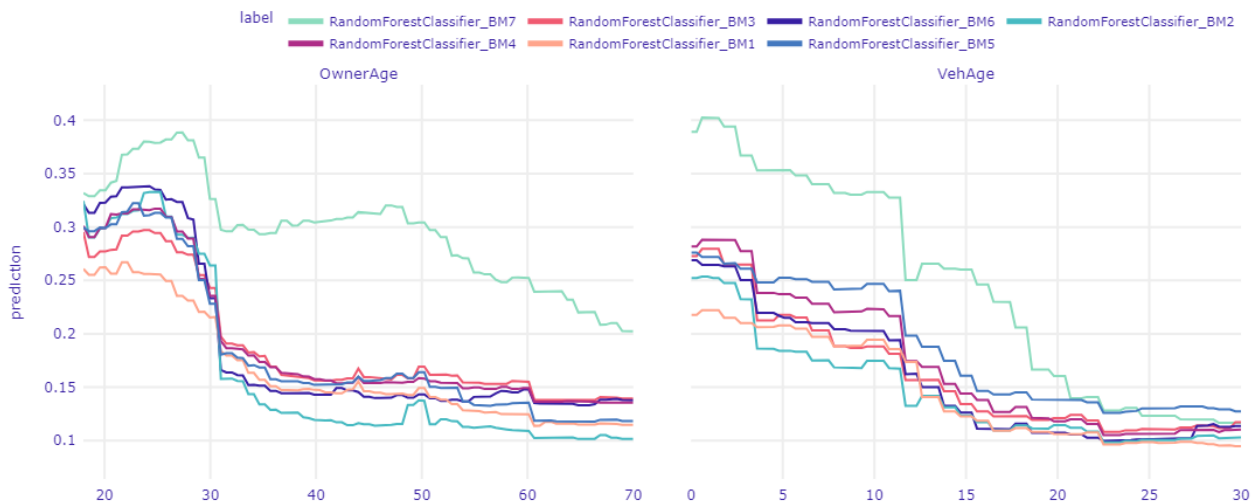


Figure 4.29: Partial Dependence profiles for the numerical data, grouped by the **BonusClass**, random forest classifier. Left panel for **OwnerAge**, right panel for **VehAge**

We can also investigate the categorical variable **BonusClass**. We can see how the prediction from the random forest classifier changes over the domain of the two numerical variables, **OwnerAge** and **VehAge**. We can see from the plots that the trend is similar for all classes, following the general form of the PD profile shown in 4.27. However, there is one interesting

class - BM7. This clearly has a higher prediction from the random forest classifier for almost all values for both **OwnerAge** and **VehAge**. We can expect this because of Figure 4.26, however, it is still surprising as supposedly BM7 is the ‘safest’ bonus class. We can see that the prediction is around the same for **OwnerAge** 18 but increases sharply and remains clearly above all of the other bonus classes, even increasing around **OwnerAge** 47 before decreasing slowly. It is interesting that the prediction remains the same from **OwnerAge** 33 to 70 for all other classes except for BM7. We can also see an intriguing trend in the **VehAge** graph; namely that the dropoff in prediction is much sharper as the prediction for **VehAge** 0 is much higher than all of the other bonus classes; peaking at 0.402 compared to a mean prediction of 0.212. The BM7 prediction steeply decreases as **VehAge** increases all the way until 20, where it joins up with the other classes and remains constant. Due to the lack of information about how the **BonusClass** data is gathered or measured, it is hard to draw real world conclusions from this, but the trends in the data are still notable and interesting.

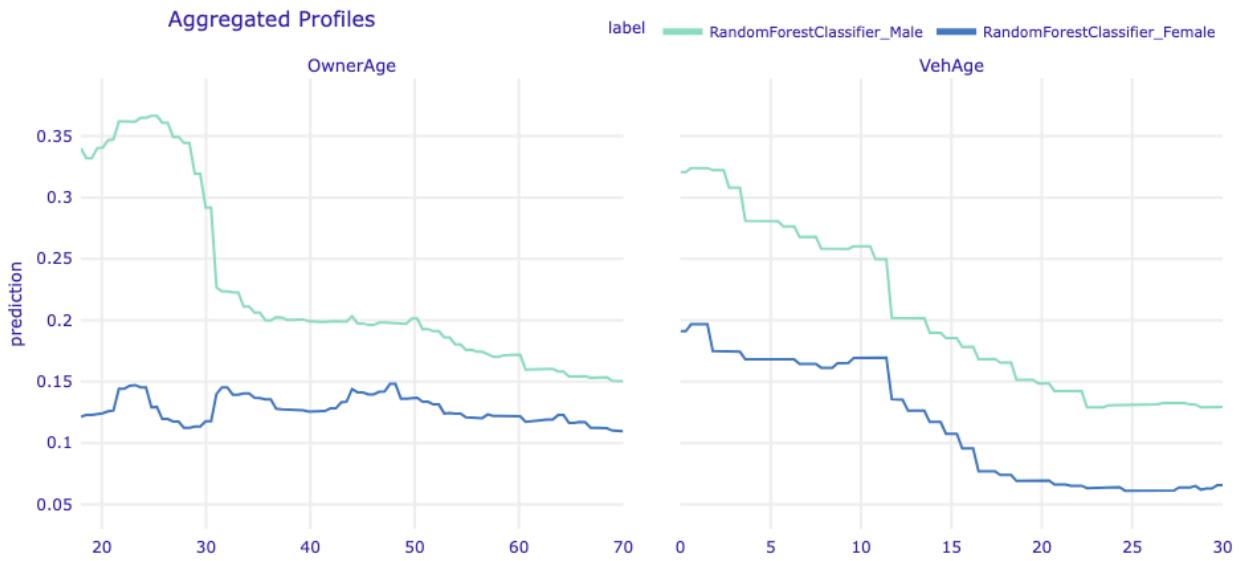


Figure 4.30: Partial Dependence profiles for the numerical data, grouped by the **Gender**, random forest classifier. Left panel for **OwnerAge**, right panel for **VehAge**

Figure 4.30 shows the partial dependence profiles for **OwnerAge** and **VehAge** split by gender. As we would expect from Figure 4.26, the male prediction is significantly higher than the female prediction for both **OwnerAge** and **VehAge**. However, the interesting thing about these graphs occurs between the ages 20 and 30 in **OwnerAge**. As we know from Figure 4.27, there is a high peak in claims made around ages 20-30 for the whole model. We also know from variable importance measures in Figure 4.25, that **OwnerAge** is the most important variable. We can now narrow down the peak in the general partial dependence profile to only men aged 20-30 as we can clearly see that the graph remains constant for women. Hence all the peaks we see in the other PD profiles around **OwnerAge** 20-30 can be attributed to men, since the **OwnerAge** PD profile is constant for women. The difference in prediction at age 25 is an extremely large 0.239. It also makes sense that even if only the men have a high prediction for ages 20-30 then it would have a big effect on the final prediction because most of the data comes from men anyway, as shown in the section 4.3. We can also see the difference in prediction for varying **VehAge**. There is a slightly steeper dropoff from **VehAge** 0 to 11 but it is minimal when compared to the differences we observed in the **OwnerAge** PD profile. These graphs suggest the majority of claims made are made by young men, especially those who have not owned a motorcycle for very long.

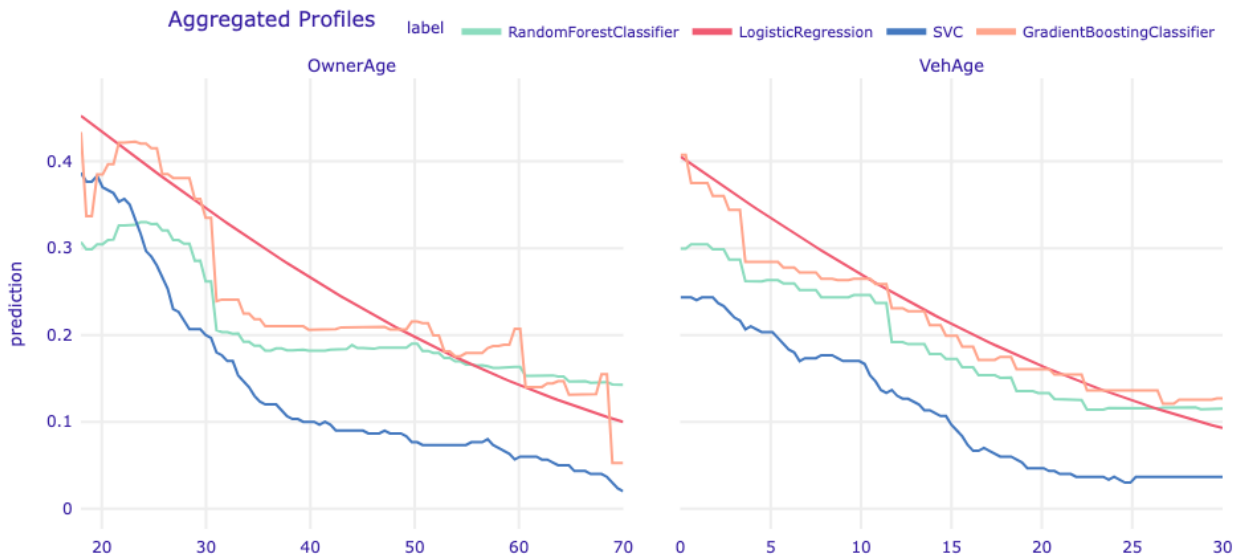


Figure 4.31: Partial Dependence profiles for all 4 models. Left panel for **OwnerAge**, right panel for **VehAge**

Figure 4.31 shows the partial dependence profiles for all 4 of the models used in this project. We can see all 4 models follow a similar pattern, with respect to the relationship between **OwnerAge**, **VehAge** and the predicted probability of a claim being made. As laid out in section 3, we will use logistic regression as a baseline to compare the other models against. The random forest and gradient boosting classifiers seem to capture the dropoff that happens at **OwnerAge** 30. This more complex relationship is something we would not have found had we only used the simpler logistic regression model. Logistic regression struggles with capturing complex non linear relationships between variables. The random forest classifier in general predicted less extreme values as the ages reach the ends of their domain. This can be expected as random forest classifiers tend make predictions that shrink towards the mean. The gradient boosting classifier fluctuates greatly at the higher end of **OwnerAge**. This could suggest overfitting as there is limited data for this section of the graph, so the gradient boosting classifier may have overfitted to the noise in the data. The SVC seems to fail to capture the peak in data that we think occurs between **OwnerAge** 20 and 30, instead predicting a steep dropoff which gradually becomes less steep. The SVC also tends to predict much lower values for both **OwnerAge** and **VehAge**. With regards to the **VehAge** profile, we can see that the logistic regression graph has managed to capture essentially the same relationship as the other models; the only differences are the shrinkage to the mean from the random forest classifier and the generally lower predictions for the SVC mentioned above.

Figure 4.32 shows the ceteris paribus (CP) profiles of 300 randomly selected motorcyclists in our dataset along with the associated partial dependence profile, which is outlined in Figure 4.27. We can see that the CP profiles for **OwnerAge** are extremely variable, but seem to follow the same pattern as the PD profile, just with different intercepts, namely a peak between ages 20-30 and a more constant prediction for the higher ages. We also see a large range of CP profiles for the **VehAge** graph, with the larger predictions having a steeper drop off to reach a similar prediction for the higher vehicle ages, suggesting confidence and lower variance as the **VehAge** increases.

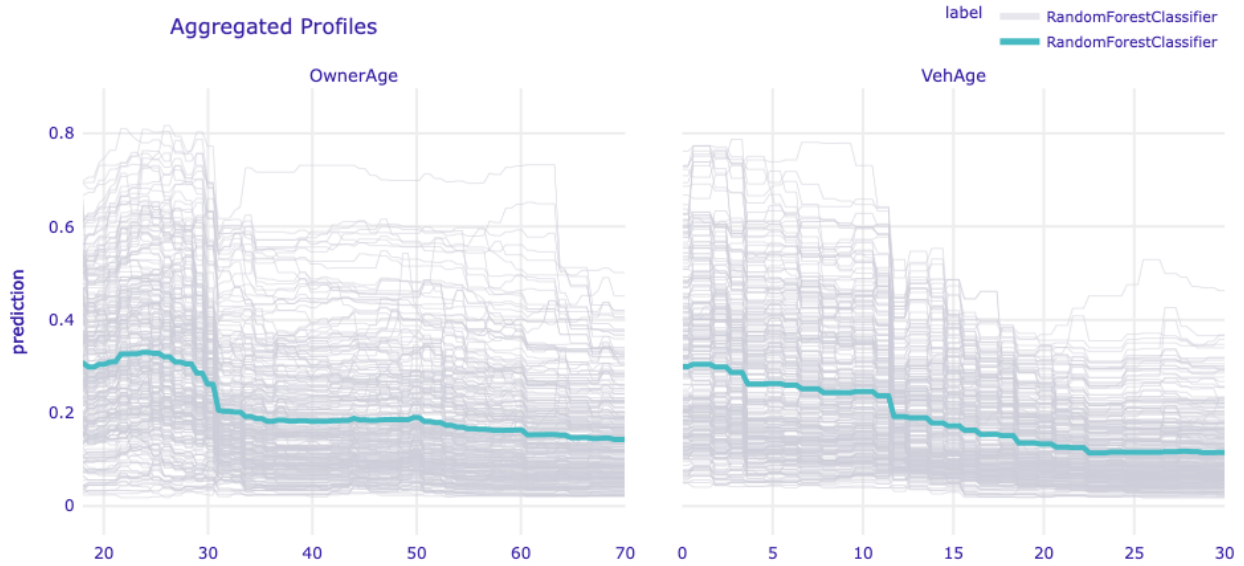


Figure 4.32: Partial Dependence profiles with corresponding CP profiles, random forest classifier

Local-dependence and Accumulated-local Profiles

We can now analyse the two numerical variables using partial dependence profiles, accumulated local effects (ALE) plots and local dependence (LD) profiles all in one graph. They each show roughly the same thing with slight differences, for example, ALE plots remove the effect of interactions between variables.

Figure 4.33 shows how the graphs follow a similar pattern to Figure 4.27. We can see that the interaction between the two variables does not have much of an effect on the final prediction because the ALE plot is almost the same as the other two profiles. This makes sense as we know the correlation between these variables is very low. We can see in the **OwnerAge** plot that the ALE plot and LD profile are almost identical until age 30 where the ALE plot is consistently above the LD and PD profiles. This could suggest a relationship between **OwnerAge** and one of the other variables is affecting the prediction for owners above the age of 30, while this is not the case for owners below 30. We can see a clear separation between the ALE plots and the LD and PD plots for **VehAge** however, suggesting a constant impact of correlation between **VehAge** and other variables. Beyond this project, one could investigate which variables are causing this disparity in the graphs we are observing, perhaps analysing the ALE plots for each **Area**, **BonusClass** or **Gender**.

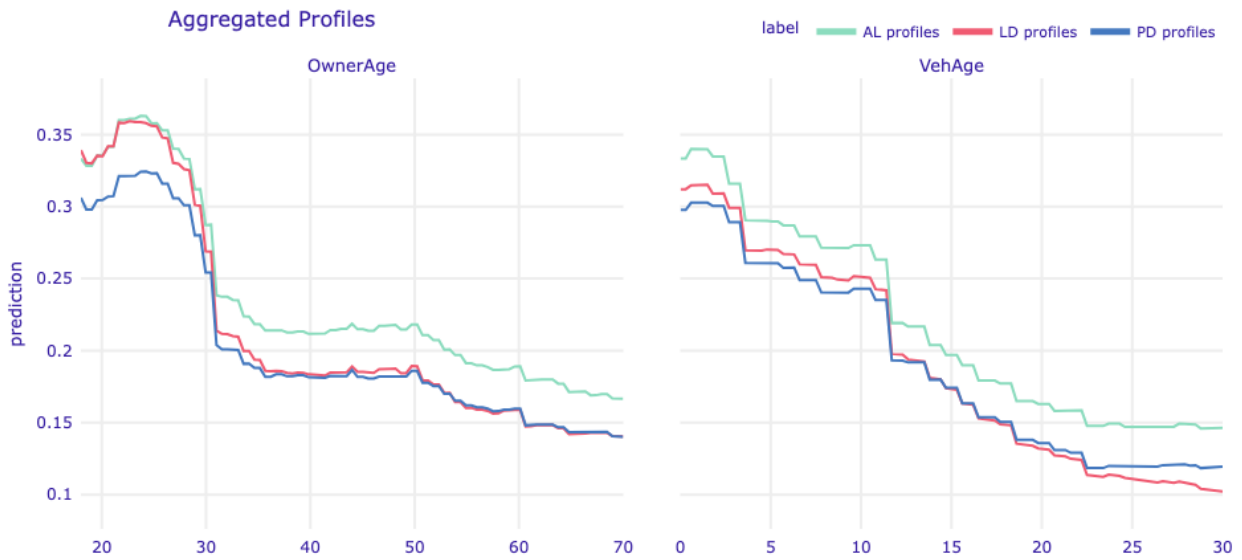


Figure 4.33: Partial Dependence, Local Dependence and Accumulated Local-Effects plots, random forest classifier

4.6 Fairness

In this section we will analyse the fairness of our modelling with regards to the **Gender** variable. We need to make sure our tools are not biased towards one **Gender** or another. It is essential that we are transparent with how we reached the conclusions we did and show that there is no bias in the tools we have used and if there is some bias we need to be aware of this. If our XAI tools have biases, it reduces the trust in them and their usability is minimised.

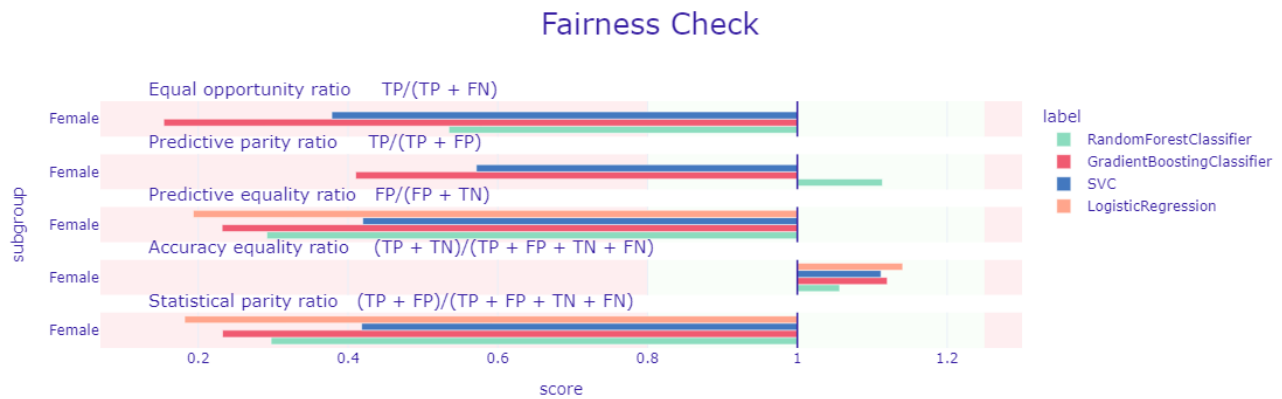


Figure 4.34: Fairness check plot

As we can see in Figure 4.34, our modelling is actually quite biased - towards women. Whenever the bar belongs to the specific model reaches the red field, it means that for this metric model is exceeding the $(\epsilon, 1/\epsilon)$ range, discussed in 3.5. So our model is less likely to predict a collision for women than men. This is partly because women have a lower rate of collision than men and partly because we have very few ‘real’ data points for women. We can see that the only fairness metric that is not violated for all models is the accuracy equality ratio. The support vector classifier and gradient boosting classifier violate all the other 4 metrics, with the logistic regression violating 2 and Random forest violating 3 bias metrics. We admit

that our modelling may show some biases but this is mostly due to a lack of data and the oversampling technique we have employed. There are various ways to reduce this bias but this is beyond the scope of the project.

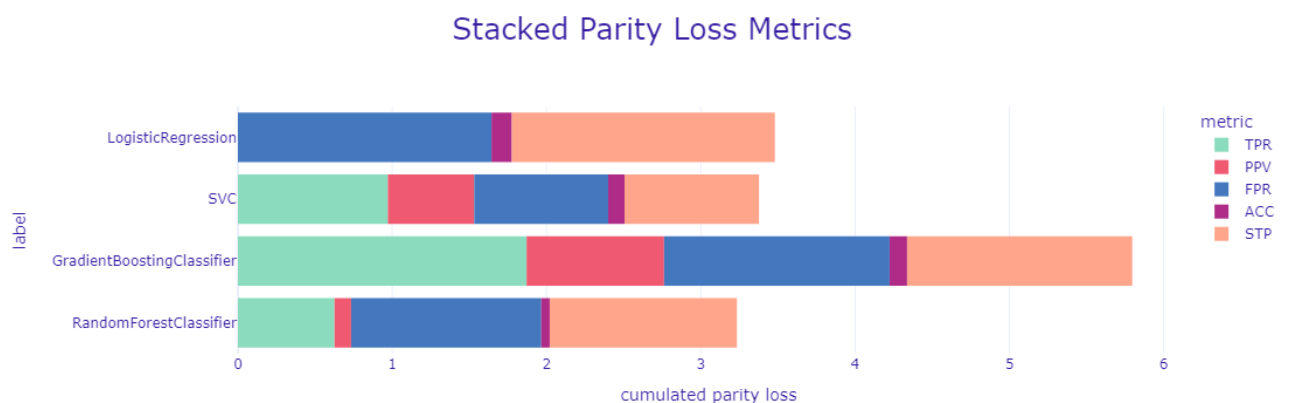


Figure 4.35: Stacked Parity Loss

Figure 4.35 shows the Stacked Parity Loss Metric, i.e. which fairness metrics contributed the most for each of our 4 models. See section 3.5 to see what the abbreviations mean. We can see that the two most consistently significant measures are the predictive equality ratio and statistical parity ratio. For SVC, GBC and random forest, the equal opportunity ratio is also a significant contributor. In terms of comparison, generally, the lower cumulative parity loss, the better the model.

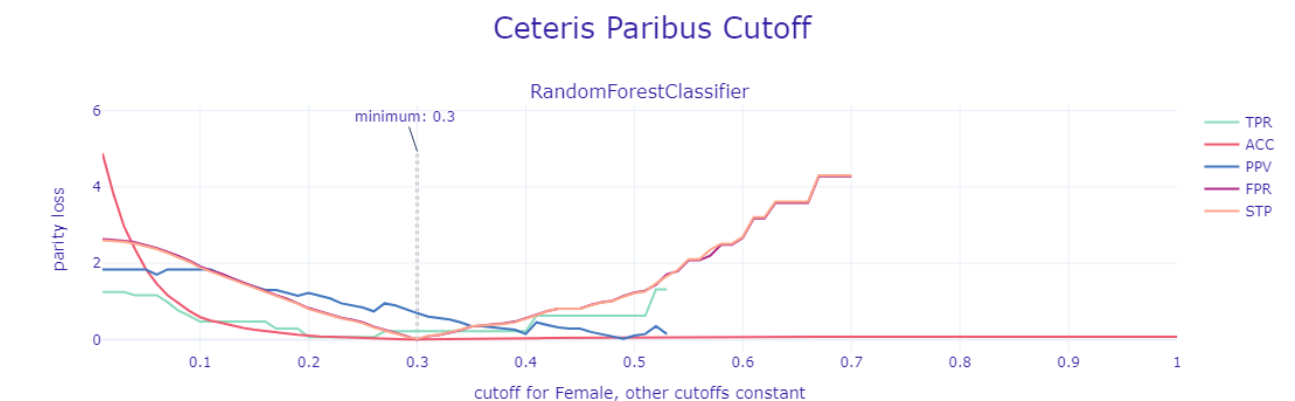


Figure 4.36: Ceteris Paribus cutoff

One way to mitigate the bias of the modelling, is to look at the CP cutoff profiles. We would then change the cutoff for Female to the minimum, to minimise the parity loss of metrics. This is one way we could reduce the bias of our model, but it is beyond the scope of this project. We can see in Figure 4.36 that the minimum parity loss is at 0.3. Additionally, such a change on the cutoff can be treated unfair since this implies lowering standards for particular subgroup.

In conclusion, we can see our models are generally biased towards women. We should be more hesitant to make conclusions using the gradient boosting classifier as **Gender** has a high impact on this model’s prediction. We should also be hesitant to use this modelling when trying to discern the effect of **Gender** as we can see our model is biased towards women. Beyond these results, it is crucial to improve the model performance and then check fairness again.

Chapter 5

Conclusion

“In many areas and sectors, interpretability will be the catalyst for the adoption of machine learning”

Christoph Molnar

Main Summary

In conclusion, XAI tools have allowed us to analyse this data set in a clear and concise manner, showing that XAI tools are ready to be used widely not only in the insurance sector, but also in other industries like healthcare, finance and retail. It is a tool which synthesises data clearly and allows for transparency and hence trustworthiness in the XAI tool. This is essential as AI becomes more prevalent in our society - we need to show that we can trust these tools.

Instance-level conclusion: From the break-down plots, we can observe the additive contribution of each explanatory variable to the model’s prediction and how they vary according to the orders. Regarding the break-down interaction plot, we can conclude that there is no interaction between explanatory variables for this single instance. Moving to Shapley values, this method tells us the average contribution of each explanatory variable, neglecting the effects of different orders. Lastly, CP profiles demonstrate the influence of different values for both numerical and categorical variables. For this selected instance, we can conclude that **Gender** and **VehAge** have the largest effects on the model’s prediction.

Dataset-level conclusions: From the variable importance measures, we can see that for all of our models, **OwnerAge** and **VehAge** are consistently the two most important variables in our model. Diving further into this analysis, we concluded that it was actually men aged 20-30 that had the most impact on our conclusions - although we know our models are biased against men so this conclusion cannot be taken in full confidence, but through other analysis it seems like this is the case. Note also that the most claims happen in the busiest cities (Zones 1-3) and also those with **BonusClass** BM7.

Limitations

One limitation of our project is the oversampling method employed. Using the ROSE method to oversample the response variable may introduce risks of overfitting and bias issues within the model. This is primarily due to the generation of duplicate data and artificial samples, which can inflate the predictive performance of the model and lead to inaccurate estimations of its generalisation capability.

Another limitation in terms of data analysis may be the lack of investigation into interaction variables. It's essential to identify potential correlations between numerical and categorical variables, as overlooking them may lead to incorrect interpretations in the explanation section.

Another significant limitation concerns the model tuning process. We relied on the default settings for training our models without adjusting any hyperparameters. This approach may result in poor model performance, as the default settings might not adequately capture the complexity of the underlying data patterns. Consequently, this could affect various aspects of the modelling process, including model selection and interpretability. Incorporating hyperparameter tuning techniques, such as grid search or randomised search, could potentially enhance the performance of our models and provide more robust and reliable results.

The bias we found in our fairness section limits the concrete conclusions about **Gender** we can make, but there are ways to mitigate this bias, for example minimising the CP cutoff in Figure 4.36. We are also frequently limited by our lack of data, we had to employ an oversampling technique to be able to create most of our graphs and tables. If we had a full set of data, this would not be necessary and the caveats that we had to outline about the lack of data in Section 4.3, 4.5.2 and 4.6 would not exist.

Further Research

Further research could involve exploring additional oversampling methods, such as SMOTE (Synthetic Minority Over-sampling Technique) and NearMiss. By experimenting with these alternative oversampling techniques, we can overcome the limitations associated with using the ROSE method, particularly the issue of high similarity among duplicate data. Additionally, we can also compare and potentially leverage other oversampling methods which might leads to improvements in model performance and accuracy.

Another further analysis can be done based on the grid search approach to fine-tune the model hyperparameters. When the default parameter setting is improved, potentially, one can expect the better model performance results. Additionally, other possible model improvements can be considered via regularisation or other tree based methods.

In addition to improving the model performances, it's also possible to implement different machine learning classification models, such as K-Nearest Neighbors (KNN) and Naive Bayes, to explore additional evaluation criteria between models.

After using grid search approach to improve the model performance, we can reply on more model performance metrics to evaluate the models, such as kappa statistics and sensitivity and specificity [Hanafy and Ming, 2021] to comprehensively understand the model performance in different dimensions.

Further research could also include deeper research into how to make our model more fair from a gender viewpoint. Firstly we could reduce the parity loss but there are other methods to reduce bias in DALEX models, namely resampling, re-weighting and ROC pivot. We could potentially employ a different method of oversampling and see how that affects the biases we found.

Bibliography

- Amina Adadi and Mohammed Berrada. Peeking inside the black-box: A survey on explainable artificial intelligence (xai). *IEEE Access*, 6:52138–52160, 2018. ISSN 2169-3536. doi: 10.1109/access.2018.2870052. URL <http://dx.doi.org/10.1109/ACCESS.2018.2870052>.
- Daniel W. Apley and Jingyu Zhu. Visualizing the effects of predictor variables in black box supervised learning models. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 82(4):1059–1086, June 2020. ISSN 1467-9868. doi: 10.1111/rssb.12377. URL <http://dx.doi.org/10.1111/rssb.12377>.
- Przemyslaw Biecek and Tomasz Burzykowski. *Explanatory Model Analysis*. Chapman and Hall/CRC, New York, 2021. ISBN 9780367135591. URL <https://pbiecek.github.io/ema/>.
- Jenna Burrell. How the machine ‘thinks’: Understanding opacity in machine learning algorithms. *Big Data and Society*, 3(1):205395171562251, January 2016. ISSN 2053-9517. doi: 10.1177/2053951715622512. URL <http://dx.doi.org/10.1177/2053951715622512>.
- Luciano Floridi, Josh Cowls, Monica Beltrametti, Raja Chatila, Patrice Chazerand, Virginia Dignum, Christoph Luetge, Robert Madelin, Ugo Pagallo, Francesca Rossi, Burkhard Schafer, Peggy Valcke, and Effy Vayena. Ai4people—an ethical framework for a good ai society: Opportunities, risks, principles, and recommendations. *Minds and Machines*, 28(4):689–707, November 2018. ISSN 1572-8641. doi: 10.1007/s11023-018-9482-5. URL <http://dx.doi.org/10.1007/s11023-018-9482-5>.
- Alicja Gosiewska and Przemyslaw Biecek. Ibreakdown: Uncertainty of model explanations for non-additive predictive models. Mar 2019.
- Mohamed Hanafy and Ruixing Ming. Machine learning approaches for auto insurance big data. *risks*, 2021. URL <https://doi.org/10.3390/risks9020042>.
- Christian Lorentzen and Michael Mayer. Peeking into the black box: An actuarial case study for interpretable machine learning. *SSRN Electronic Journal*, 2020. ISSN 1556-5068. doi: 10.2139/ssrn.3595944. URL <http://dx.doi.org/10.2139/ssrn.3595944>.
- G. Lunardon, N. Menardi and N. Torelli. Rose:a package for binary imbalanced learning. *The R Journal*, 6:79, 2014. doi: 10.32614/rj-2014-008. URL <https://doi.org/10.32614/rj-2014-008>.
- Michael Mayer, Daniel Meier, and Mario V. Wuthrich. Shap for actuaries: Explain any model. *SSRN Electronic Journal*, 2023. ISSN 1556-5068. doi: 10.2139/ssrn.4389797. URL <http://dx.doi.org/10.2139/ssrn.4389797>.
- Trevor Maynard, Luca Baldassarre, Yves-Alexandre de Montjoye, Liz McFall, and María Óskarsdóttir. Ai: Coming of age? *Annals of Actuarial Science*, 16(1):1–5, January 2022. ISSN 1748-5002. doi: 10.1017/s1748499521000245. URL <http://dx.doi.org/10.1017/S1748499521000245>.

- Ugochukwu Orji and Elochukwu Ukwandu. Machine learning for an explainable cost prediction of medical insurance. *Machine Learning with Applications*, 15:100516, March 2024. ISSN 2666-8270. doi: 10.1016/j.mlwa.2023.100516. URL <http://dx.doi.org/10.1016/j.mlwa.2023.100516>.
- Emer Owens, Barry Sheehan, Martin Mullins, Martin Cunneen, Julianne Ressel, and German Castignani. Explainable artificial intelligence (xai) in insurance. *Risks*, 10(12):230, December 2022. ISSN 2227-9091. doi: 10.3390/risks10120230. URL <http://dx.doi.org/10.3390/risks10120230>.
- Wei Qiu, Hugh Chen, Ayse Berceste Dincer, Scott Lundberg, Matt Kaeberlein, and Su-In Lee. Interpretable machine learning prediction of all-cause mortality. *Communications Medicine*, 2(1), October 2022. ISSN 2730-664X. doi: 10.1038/s43856-022-00180-x. URL <http://dx.doi.org/10.1038/s43856-022-00180-x>.
- Tarid Wongvorachan, Surina He, and Okan Bulut. A comparison of undersampling, oversampling, and smote methods for dealing with imbalanced classification in educational data mining. *Information*, 14:54, 01 2023. doi: 10.3390/info14010054.
- Mario V. Wüthrich and Michael Merz. *Statistical Foundations of Actuarial Learning and its Applications*. Springer International Publishing, 2023. ISBN 9783031124099. doi: 10.1007/978-3-031-12409-9. URL <http://dx.doi.org/10.1007/978-3-031-12409-9>.

Appendix A

Code

Code availability: <https://github.com/RichardHuang24/XAI-Mathematics-Project>

Appendix B

Instance-level

We have included plots for both instance-level and dataset-level generated by the gradient boosting classifier to facilitate comparison with the results of the random forest classifier.

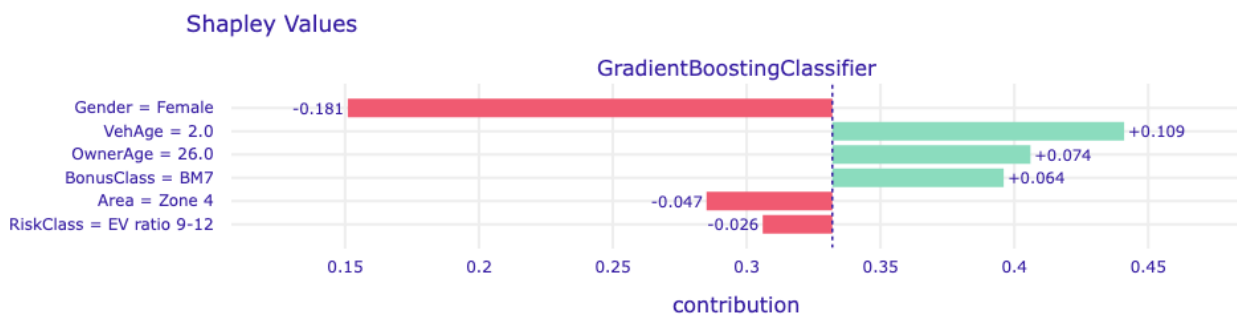


Figure B.1: Gradient boosting classifier Shapley values plot

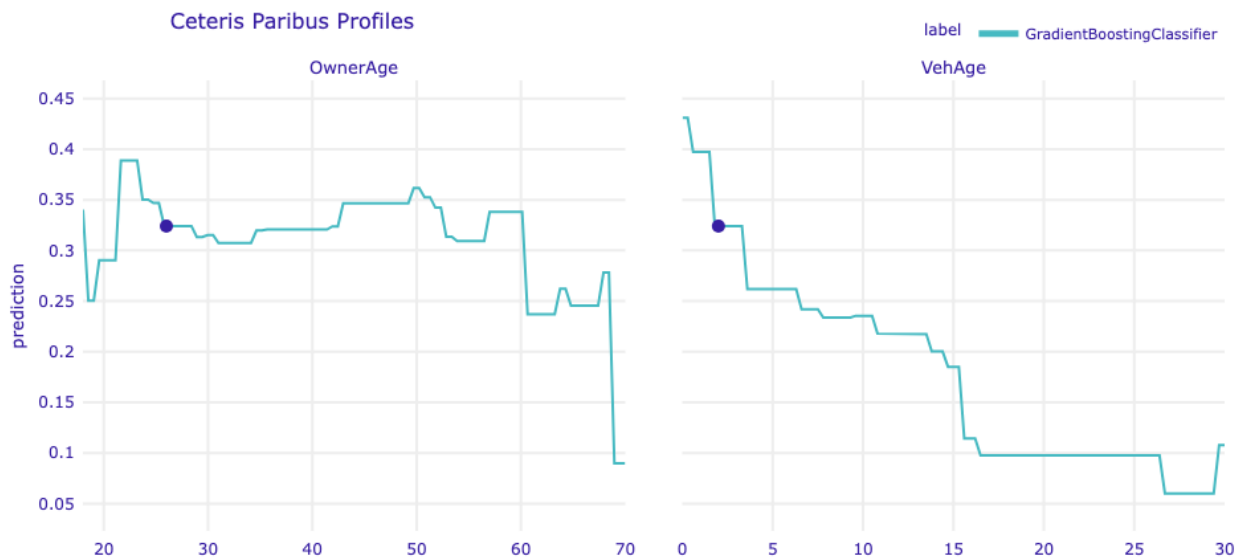


Figure B.2: Gradient boosting classifier CP profiles numerical variables

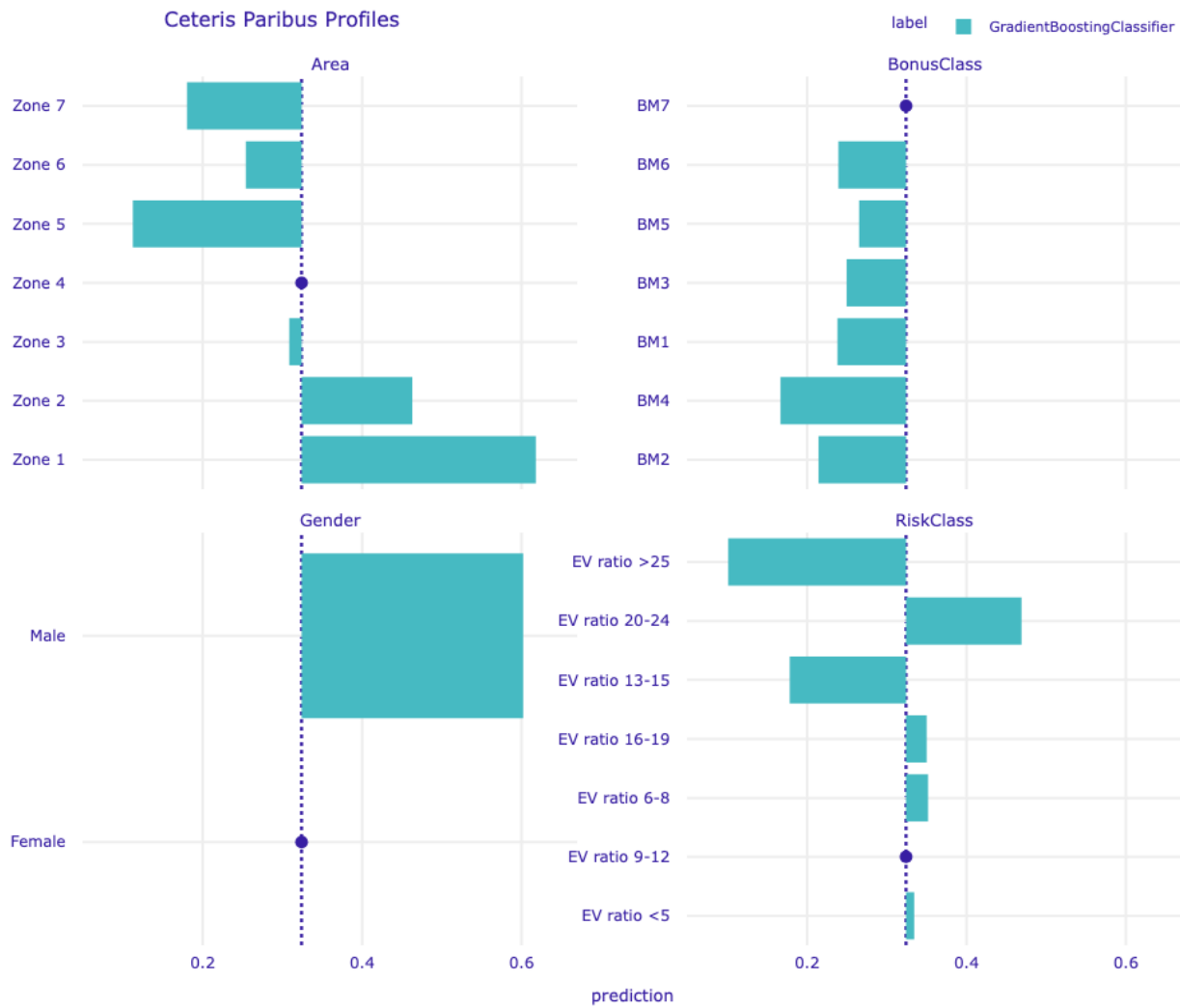


Figure B.3: Gradient boosting classifier CP profiles categorical variables

Appendix C

Dataset-level

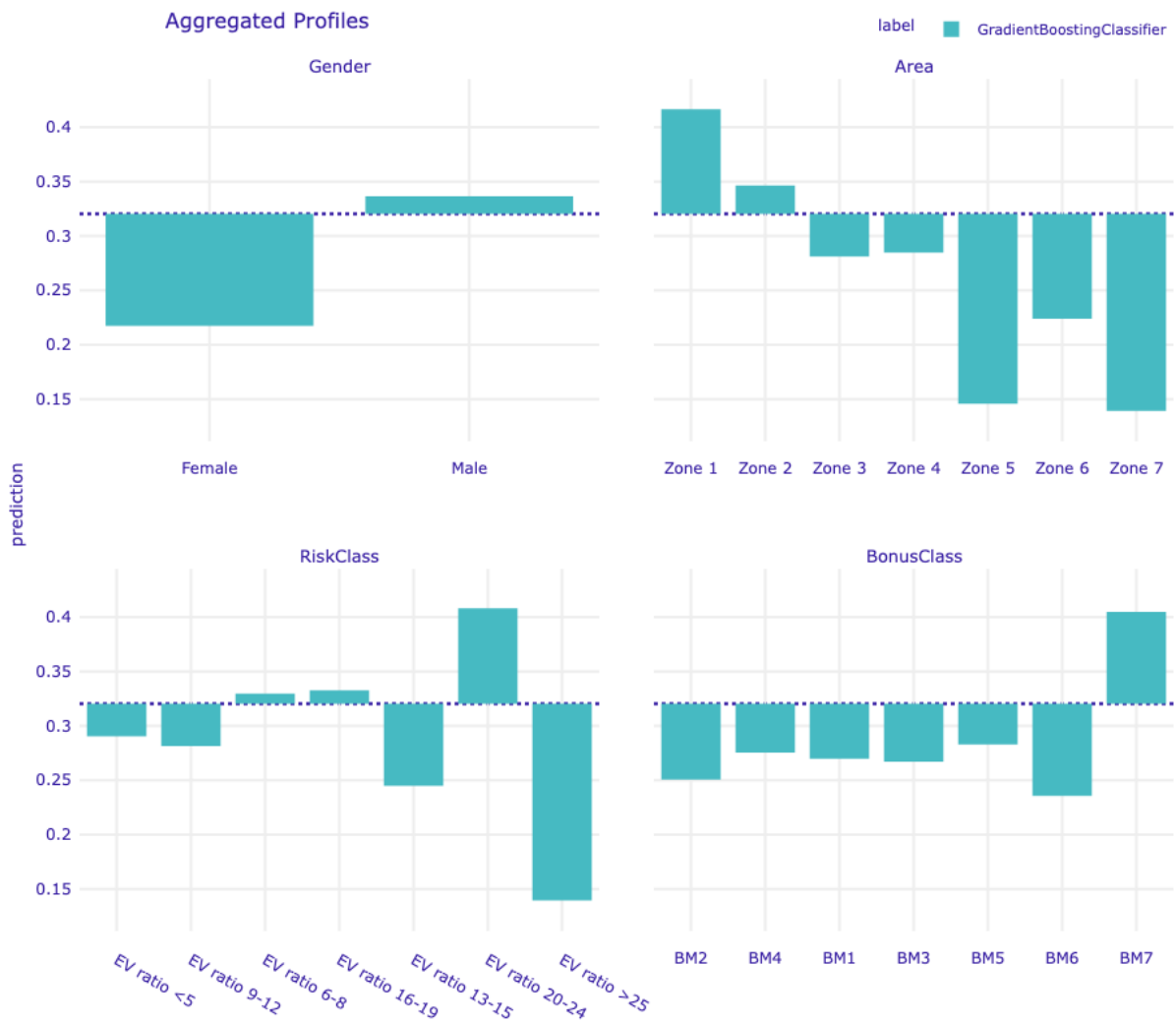


Figure C.1: Gradient boosting classifier Partial Dependence profiles categorical variables

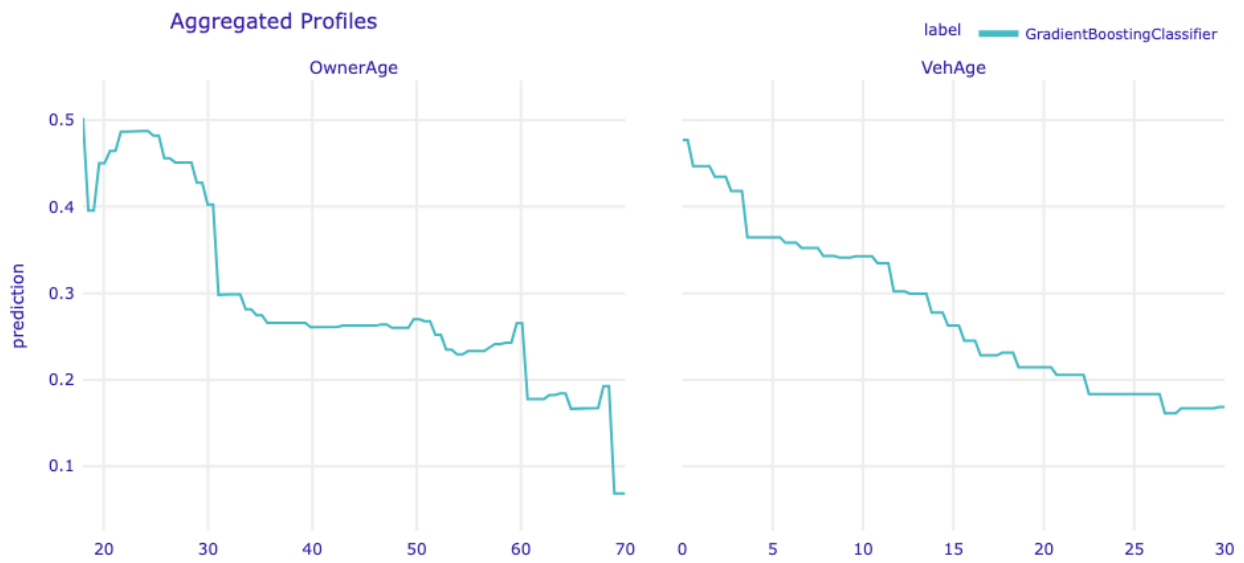


Figure C.2: Gradient boosting classifier Partial Dependence profiles numerical variables

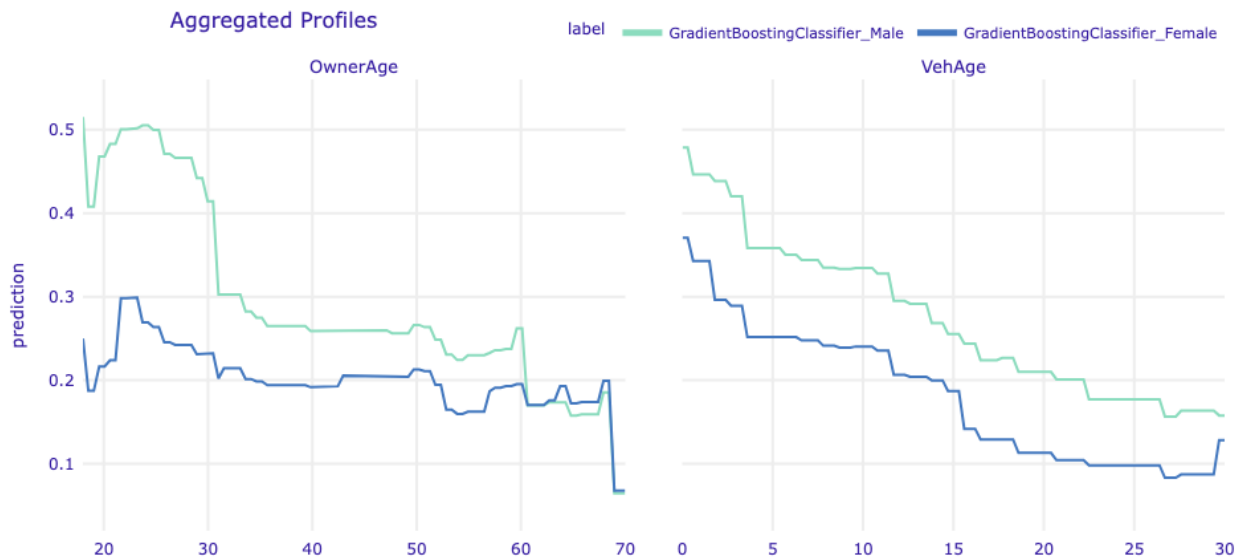


Figure C.3: Gradient boosting classifier Partial Dependence profiles numerical variables, grouped by the Gender